



**Singapore
Institute of
Management**



**UNIVERSITY
OF WOLLONGONG
AUSTRALIA**

CSIT-26-S3-30: AI-Based Misinformation Detection for Short-Form Content

FYP-26-S3-30: Technical Design Manual

Team

Student Name	UOW ID	Student E-mail
Ho Sze Wei, Donovan	8949724	swdho001@mymail.sim.edu.sg
Wong Yi Da	1037274	ydwong003@mymail.sim.edu.sg
Matthew Alexander Peeris	1052585	peeris001@mymail.sim.edu.sg
CHU WAI CHUNG	8949256	wcchu001@mymail.sim.edu.sg
POON CHUN PING	8865747	cppoon001@mymail.sim.edu.sg

Supervisor:

Mr Liaw Chun Huei

Table of Contents

1. Introduction	1
2. Purpose	1
3. Design Approach	2
3.1 System Environment	2
3.2 Data Flow	3
3.3 Architecture Design	4
4. Activity Diagrams	5
4.1 System Administrator	5
4.2 Free User	6
4.3 Premium User	7
4.4 Data Engineer	8
5. Sequence Diagrams.....	9
5.1 System Administrator	9
5.2 Free User	20
5.3 Premium User	37
5.4 Data Engineer	45
6. User Interface	53
6.1 Login and Account Creation.....	53
6.2 Free User	54
6.3 Premium User	56
6.4 System Administrator	58
6.5 Data Engineer	60
7. Database Design	62
Mermaid Diagram	62
7.1. Users Collection	63
7.2. Subscriptions Collection	64
7.3. Usage Allowances Collection.....	64
7.4. Analysis Results Collection.....	65
7.5. Incorrect Result Reports Collection.....	66
7.6. Feedback Collection	66
7.7. Sources Collection.....	66
7.8. Pipeline Runs Collection.....	67

7.9. Audit Logs Collection	67
7.10. Supporting Firebase Services	68

1. Introduction

The objective of this document is to give an overview of the technical design for an AI-Based Misinformation Detection for Short-Form Content

2. Purpose

The purpose of the Technical Design Manual (TDM) is to provide a comprehensive overview of the architecture, components, and implementation details for our application. This document would serve as a reference for our team to be consistent and clear with our design and development of the system.

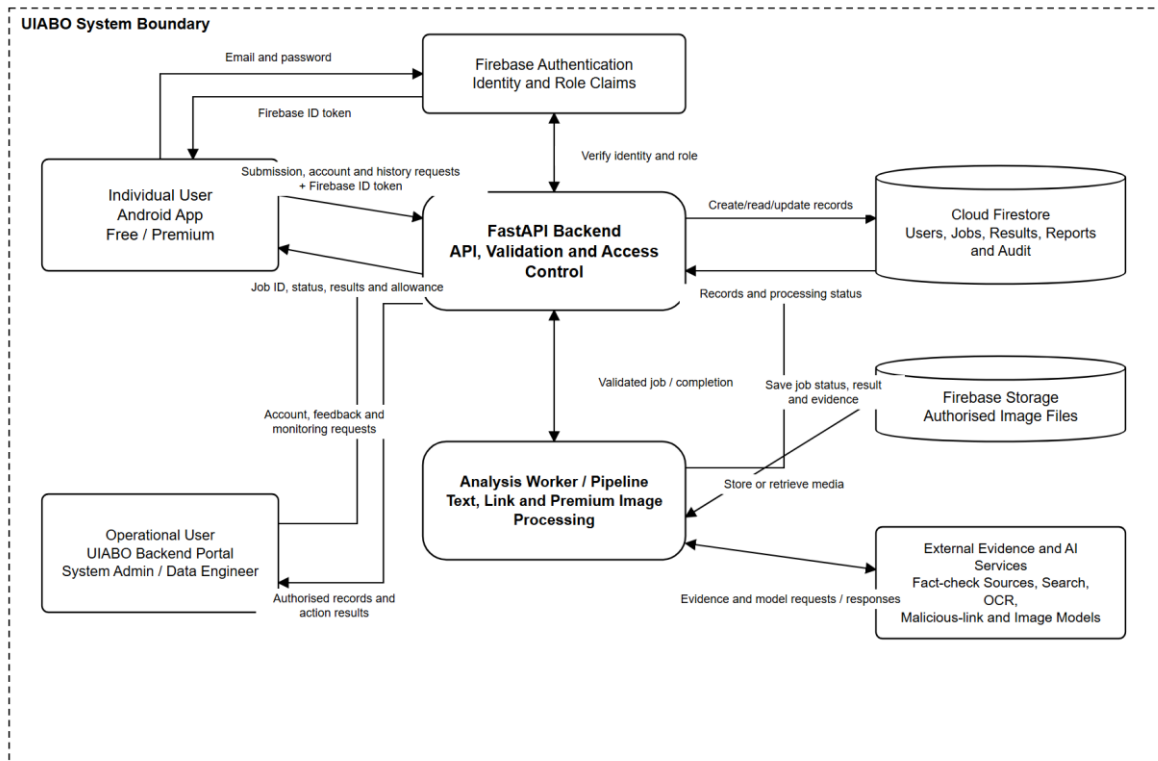
3. Design Approach

3.1 System Environment

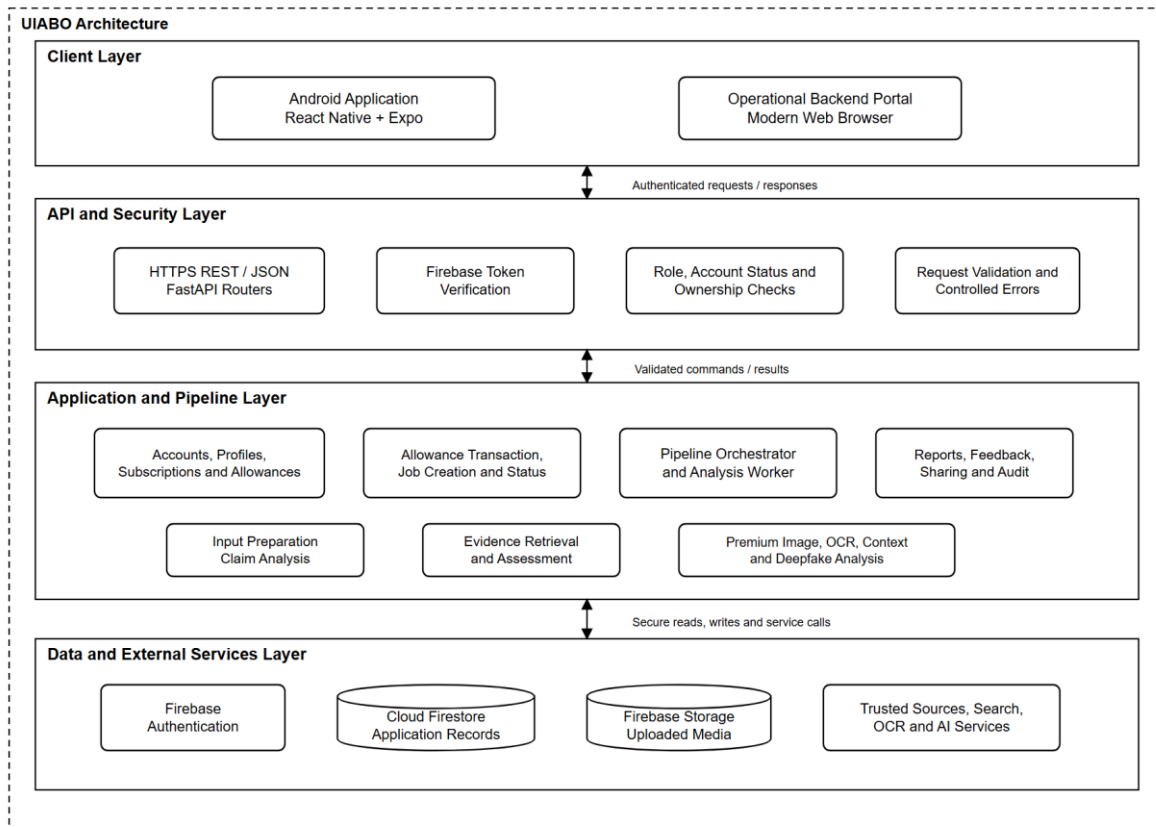
UIABO uses a separated client, API, analysis-pipeline and Firebase architecture so the Android application and operational portal do not contain secret service credentials or AI processing logic.

- Consumer client: Android smartphone running the React Native application built with Expo.
- Operational client: UIABO Backend Portal accessed through a modern Chrome, Edge or Firefox browser.
- Back end: Python with FastAPI and Uvicorn for authenticated REST API services.
- Communication: HTTPS requests and JSON responses; the Android application uses Axios.
- Firebase services: Firebase Authentication, Cloud Firestore and Firebase Storage.
- Analysis environment: input preparation, claim analysis, evidence retrieval, evidence assessment and premium image-processing components.
- Development and testing: Visual Studio Code, Expo, Android Studio Emulator, Postman and Firebase Console.
- Network requirement: Internet access to the UIABO API, Firebase and approved evidence or AI services.

3.2 Data Flow

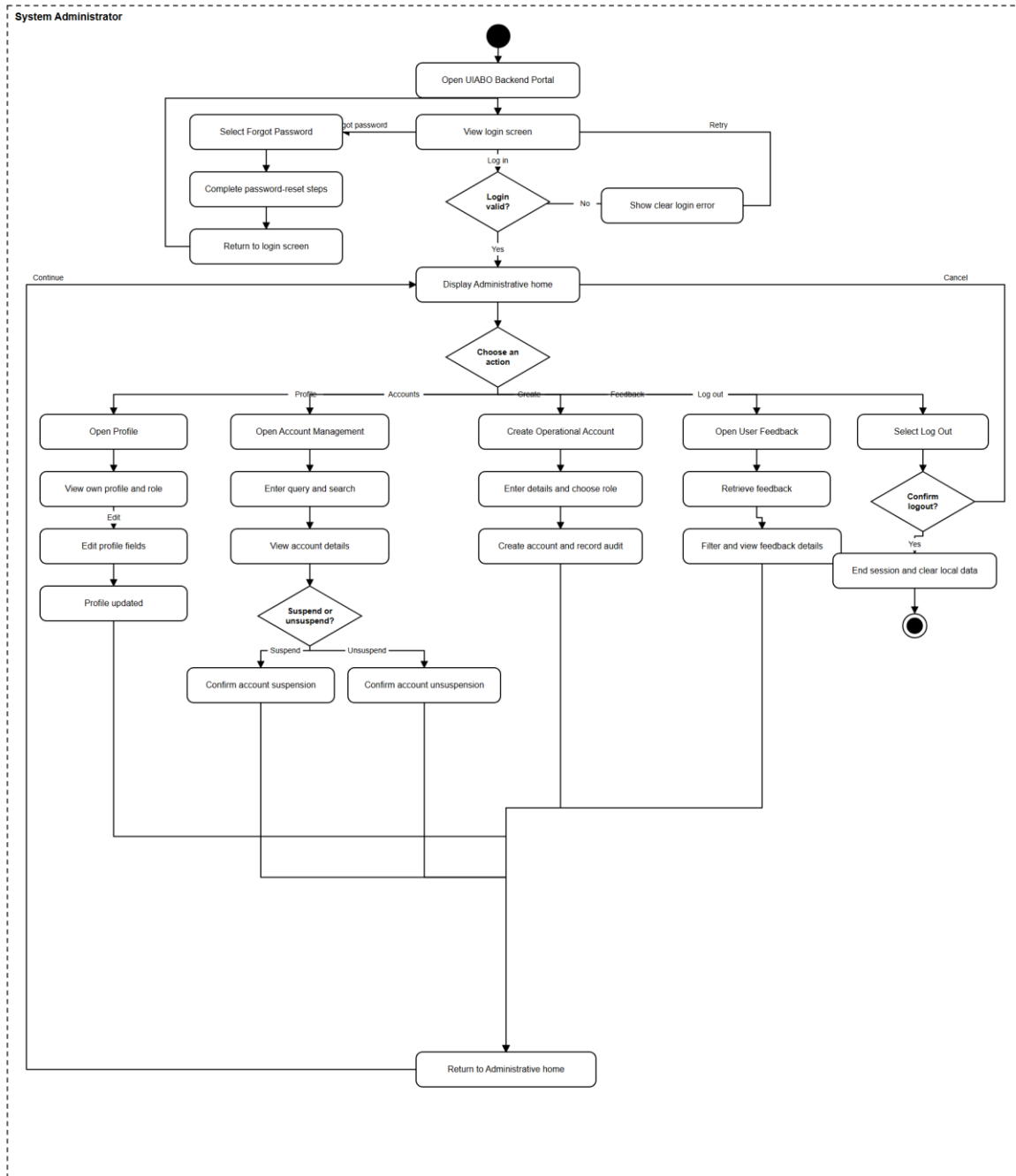


3.3 Architecture Design

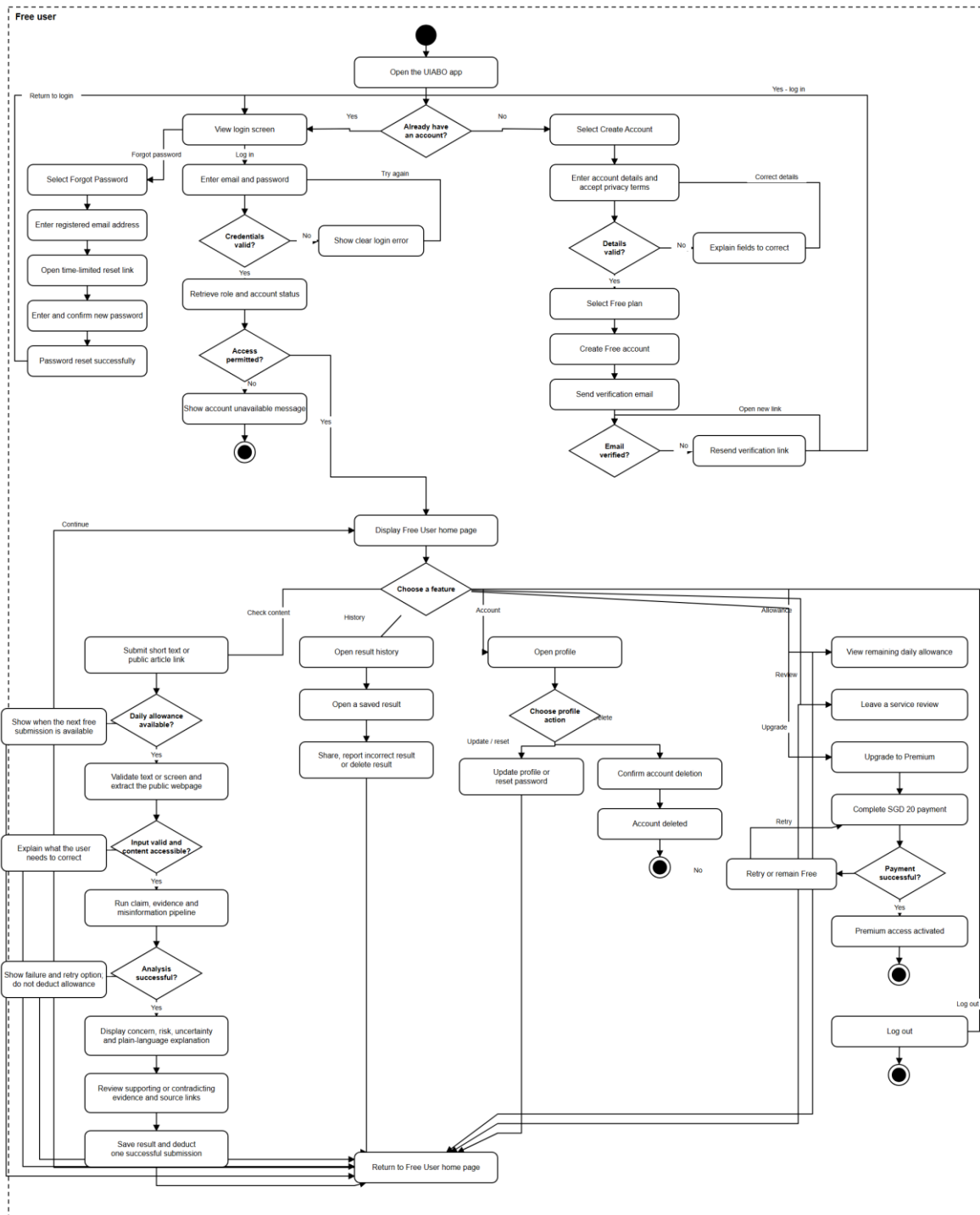


4. Activity Diagrams

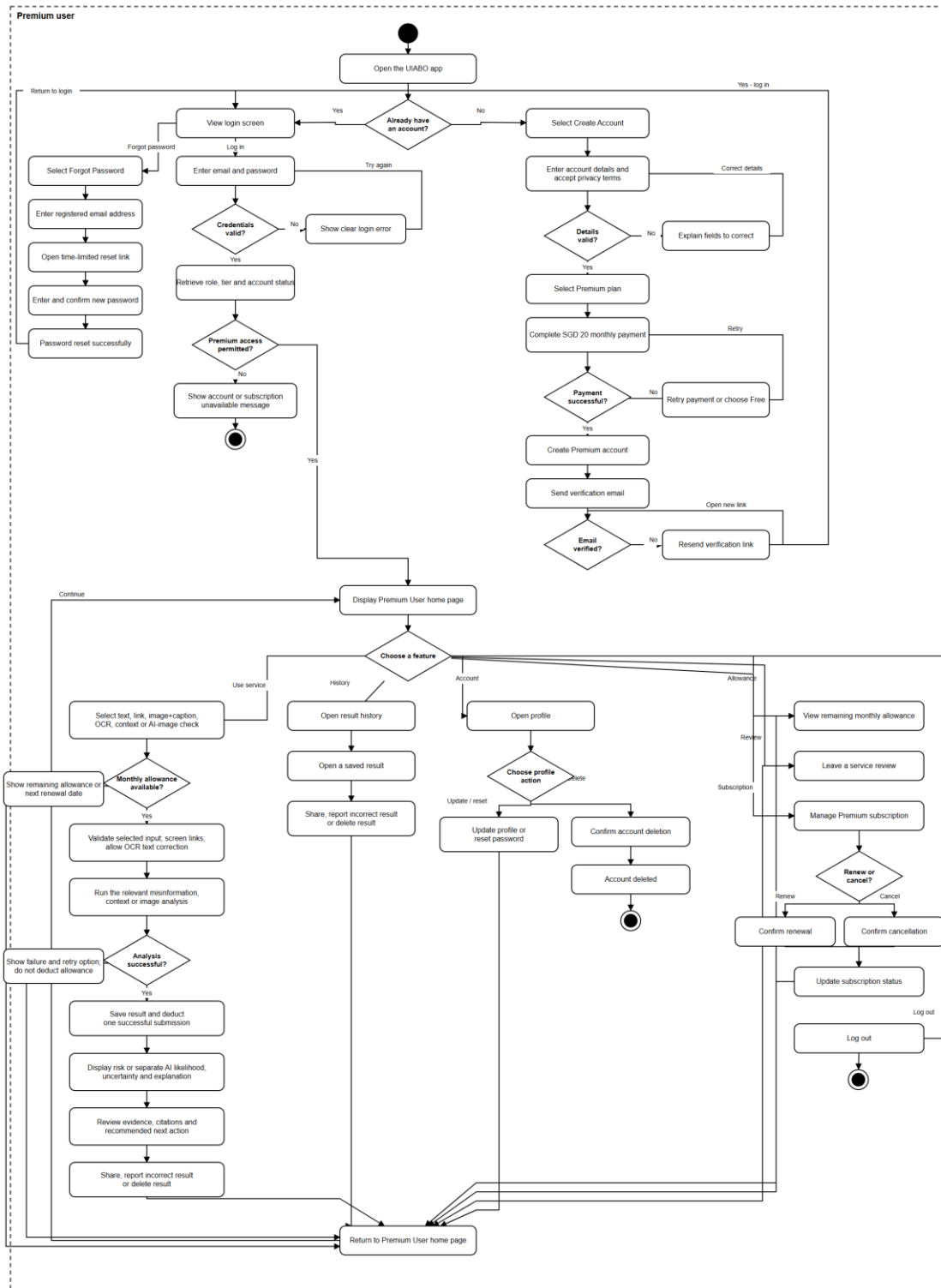
4.1 System Administrator



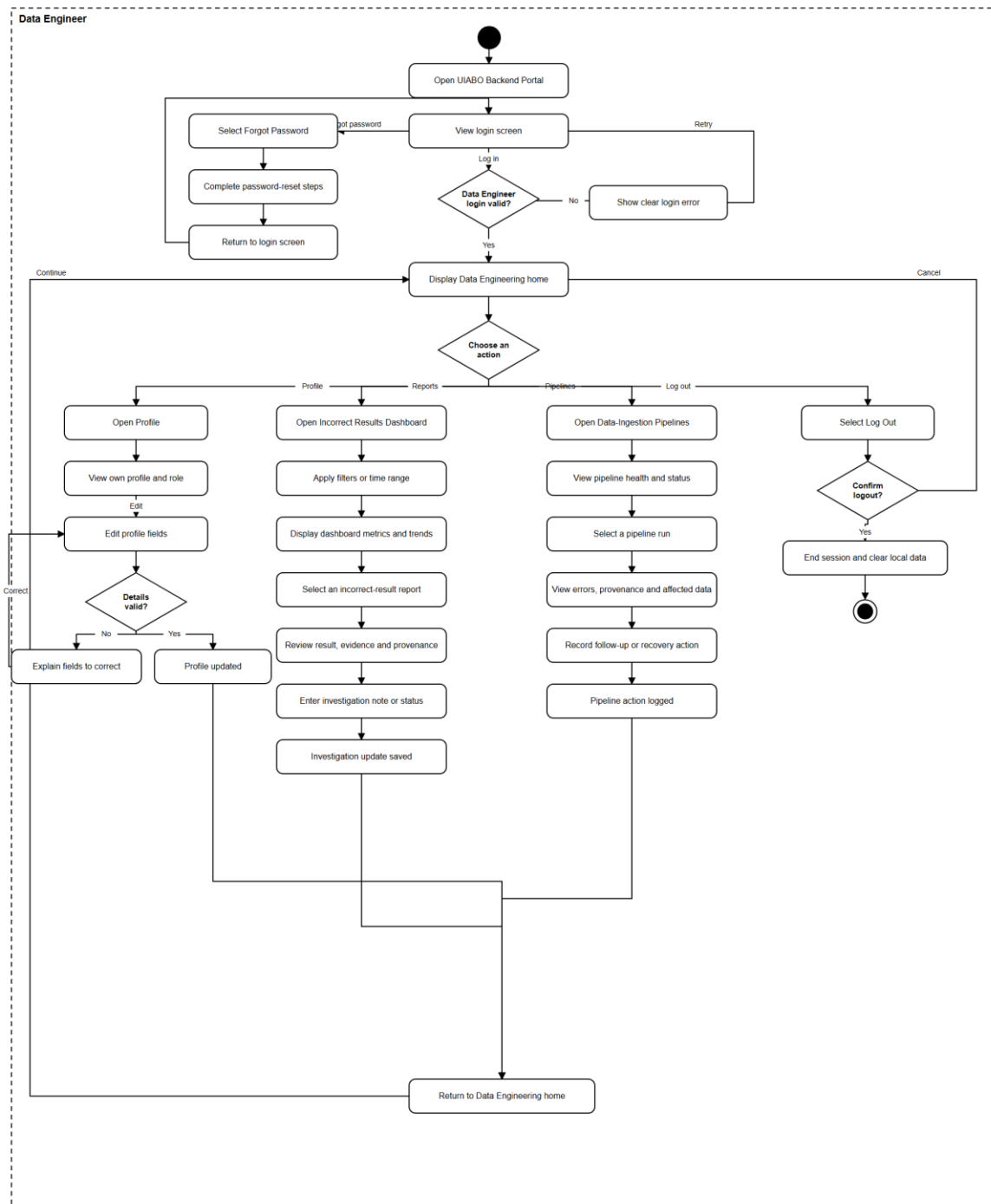
4.2 Free User



4.3 Premium User



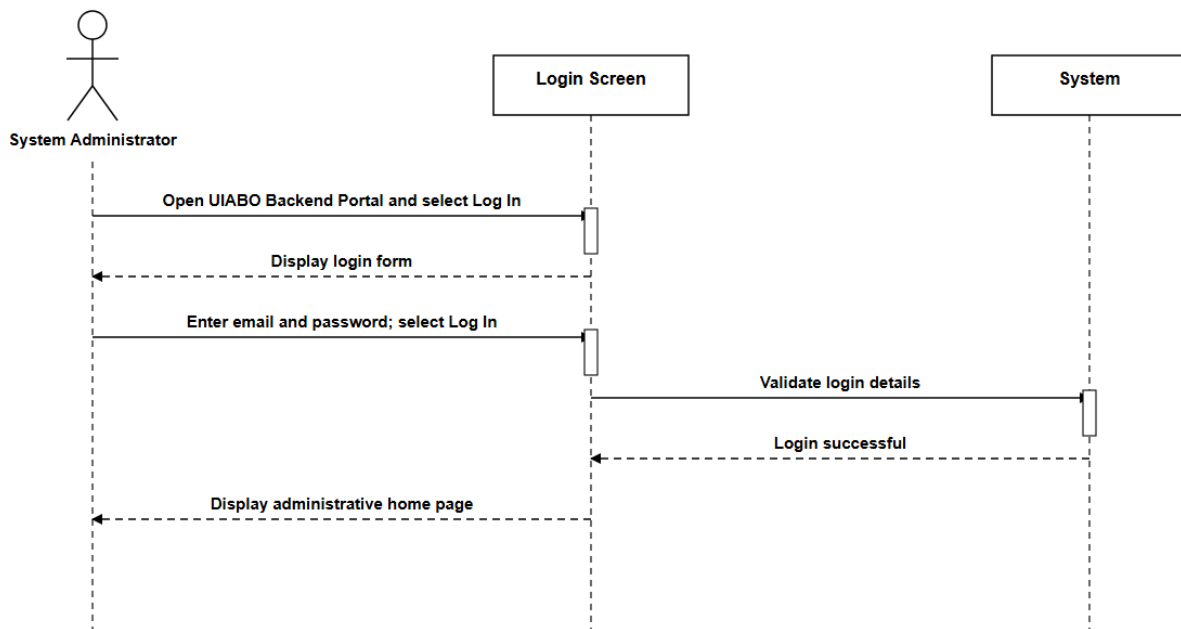
4.4 Data Engineer



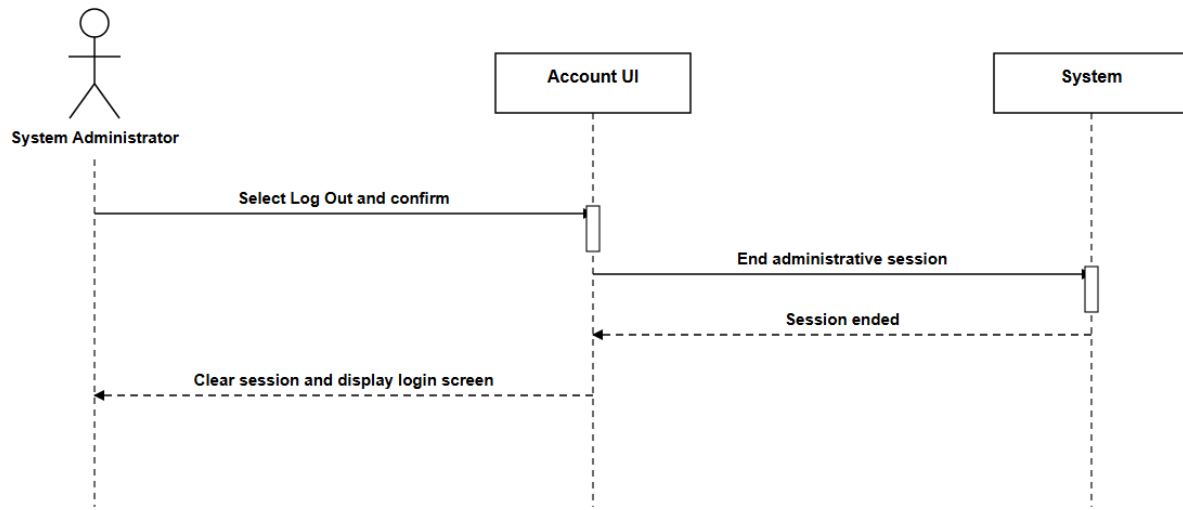
5. Sequence Diagrams

5.1 System Administrator

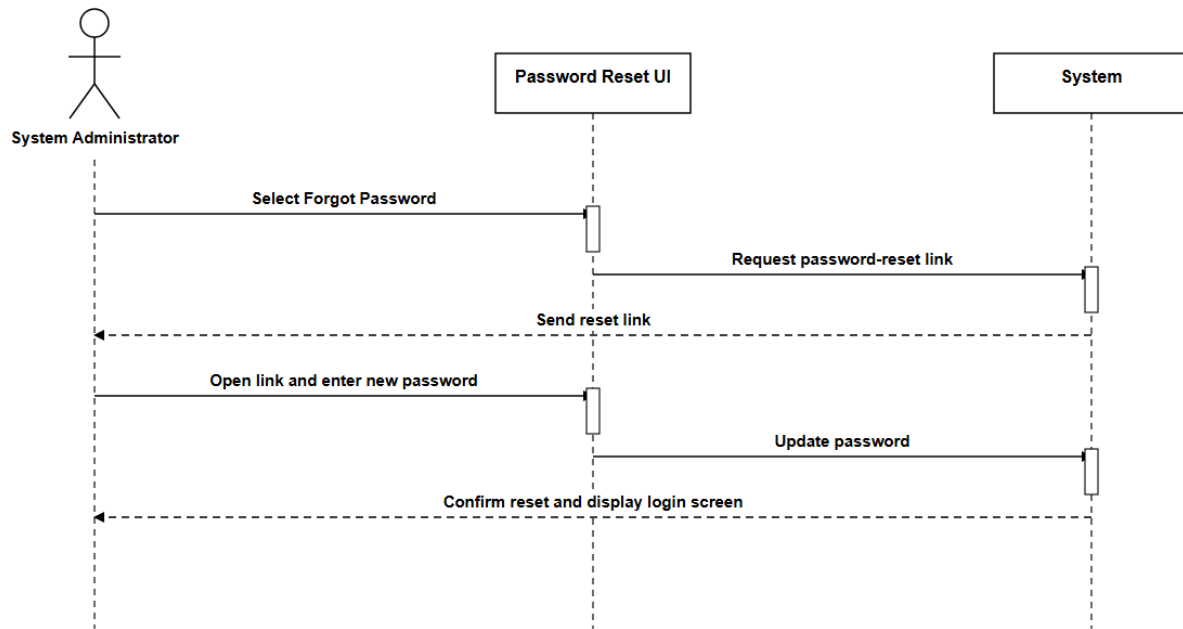
Login (System Administrator)



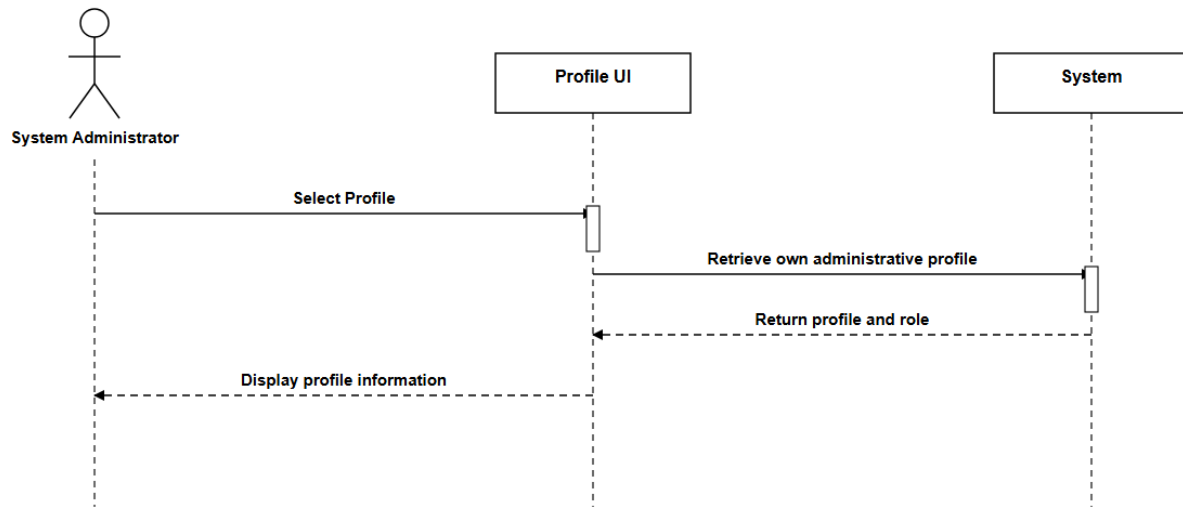
Logout (System Administrator)



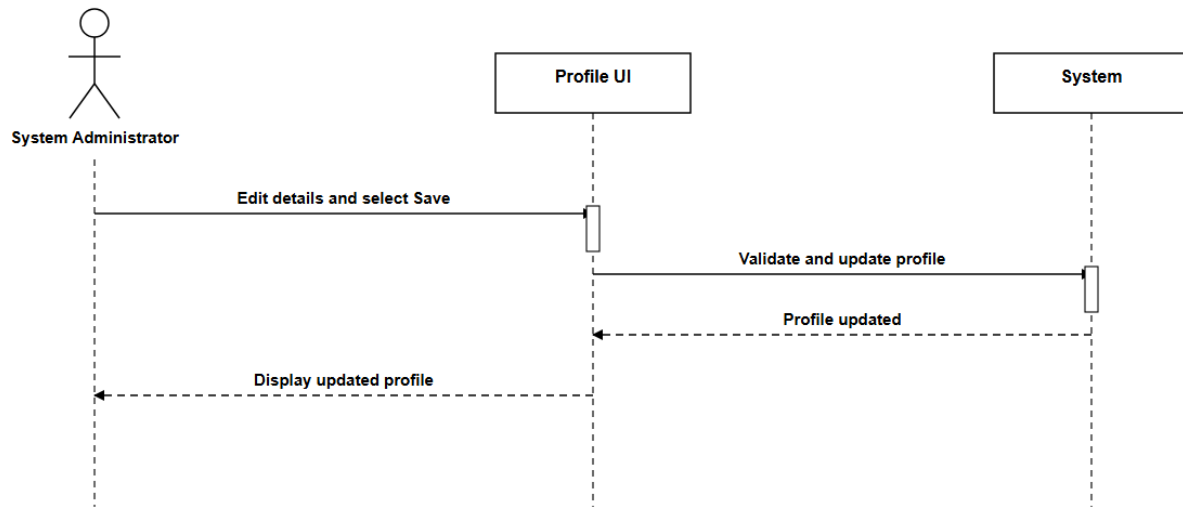
Reset Password (System Administrator)



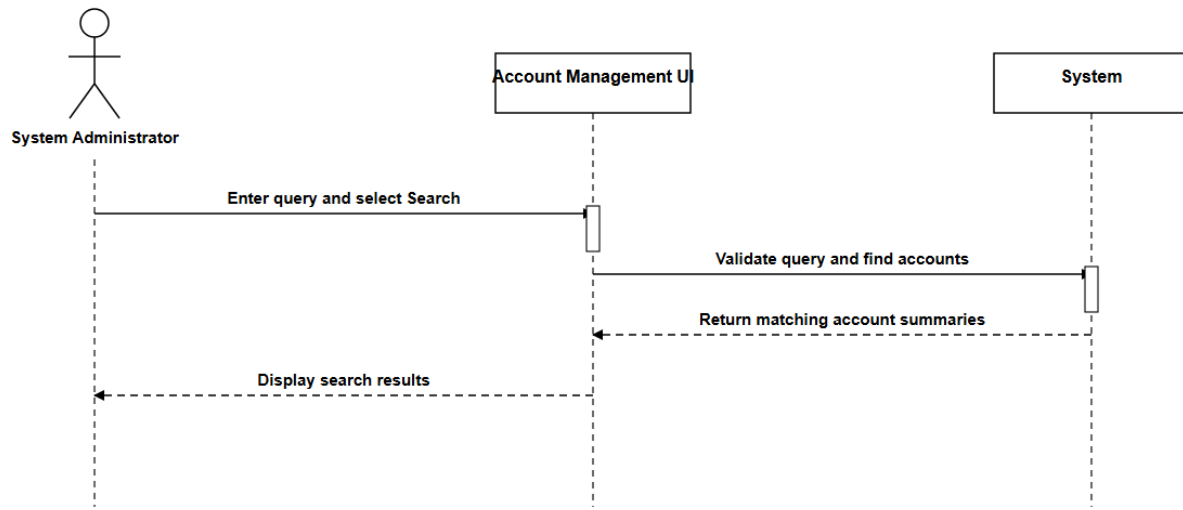
View Profile (System Administrator)



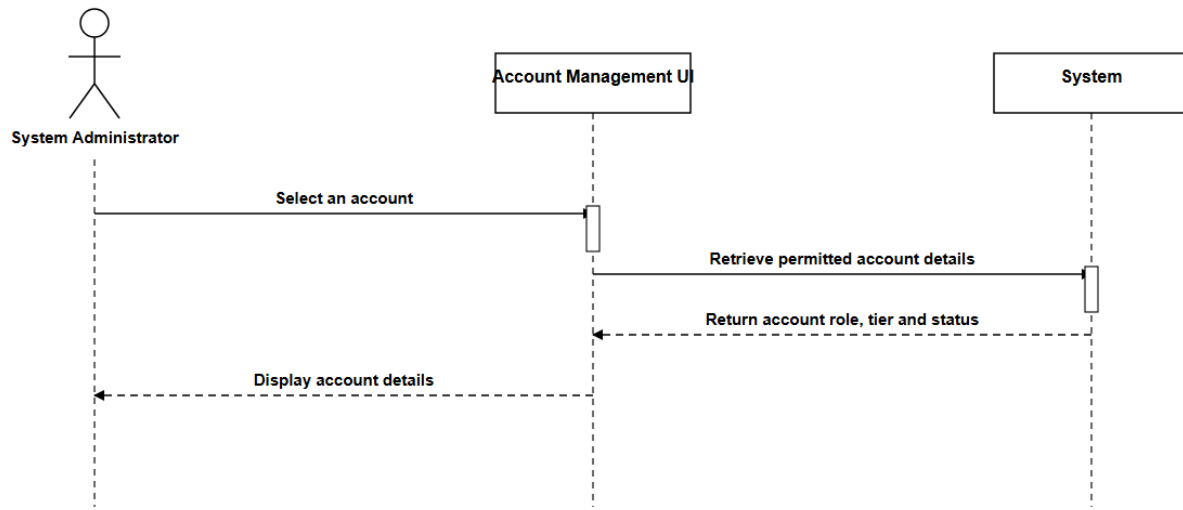
Update Profile (System Administrator)



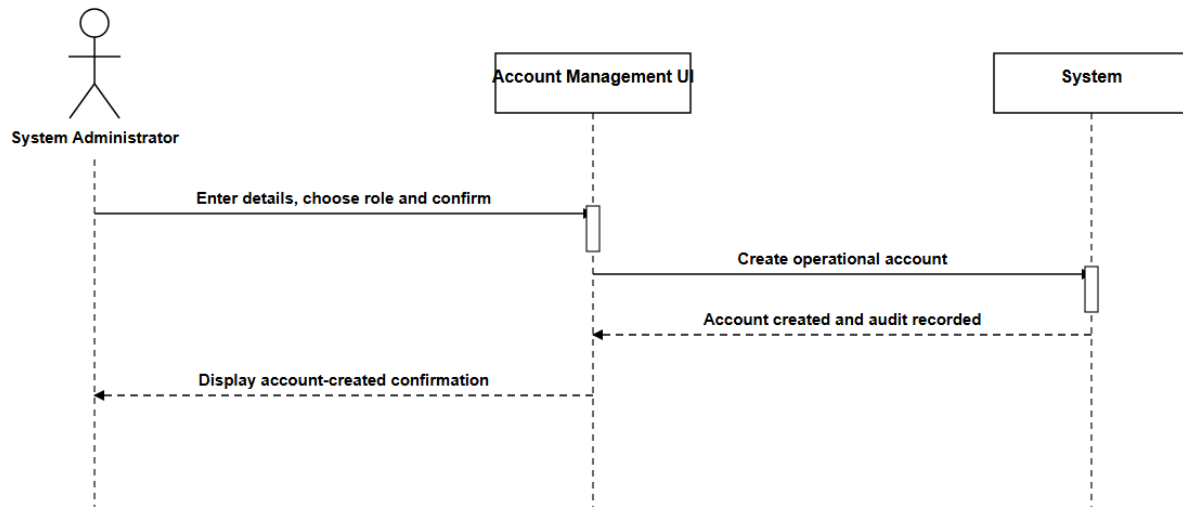
Query Account (System Administrator)



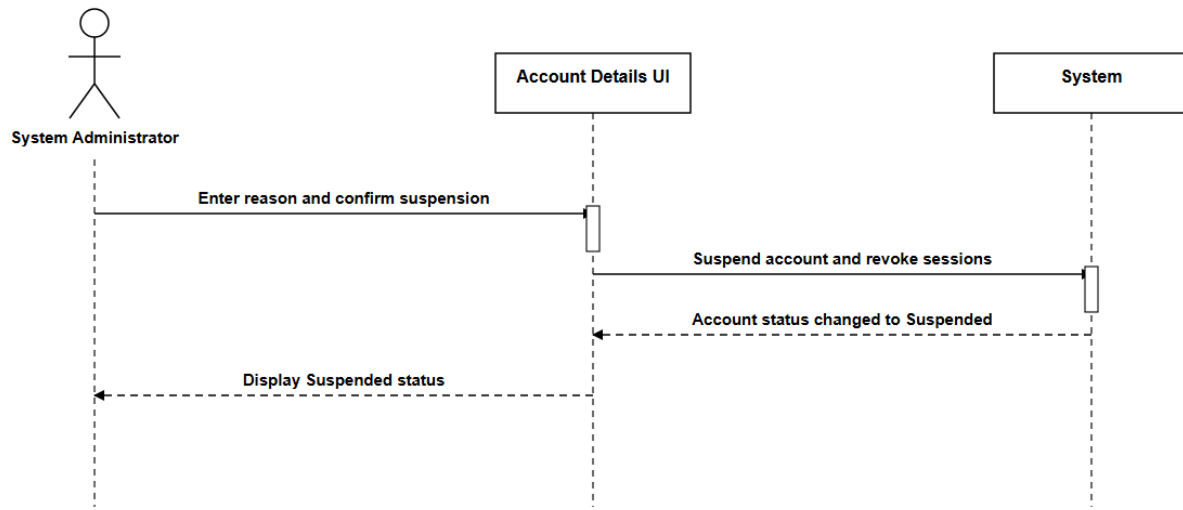
View Account Details (System Administrator)



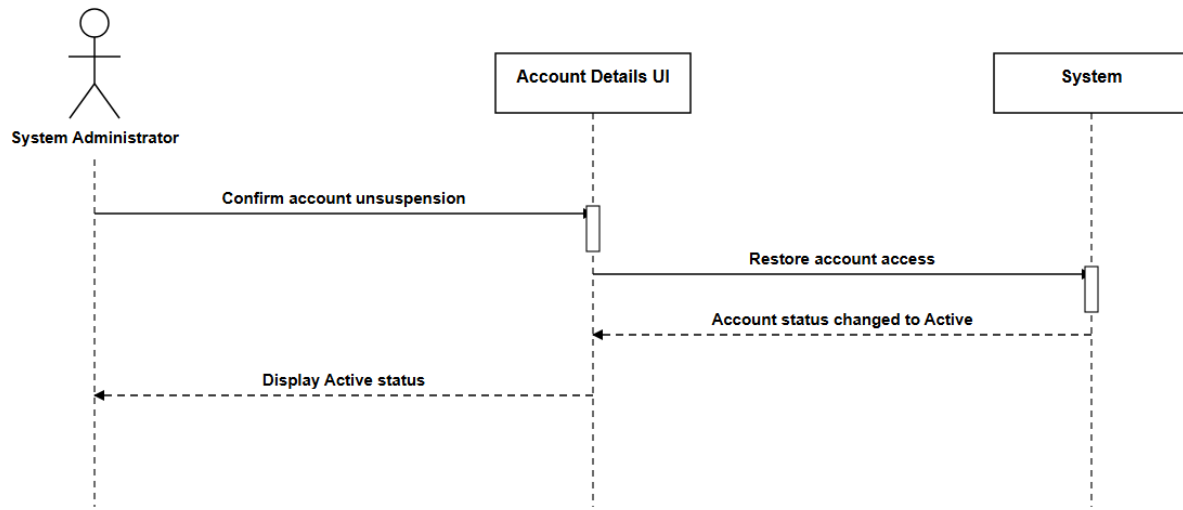
Create Operational Account (System Administrator)



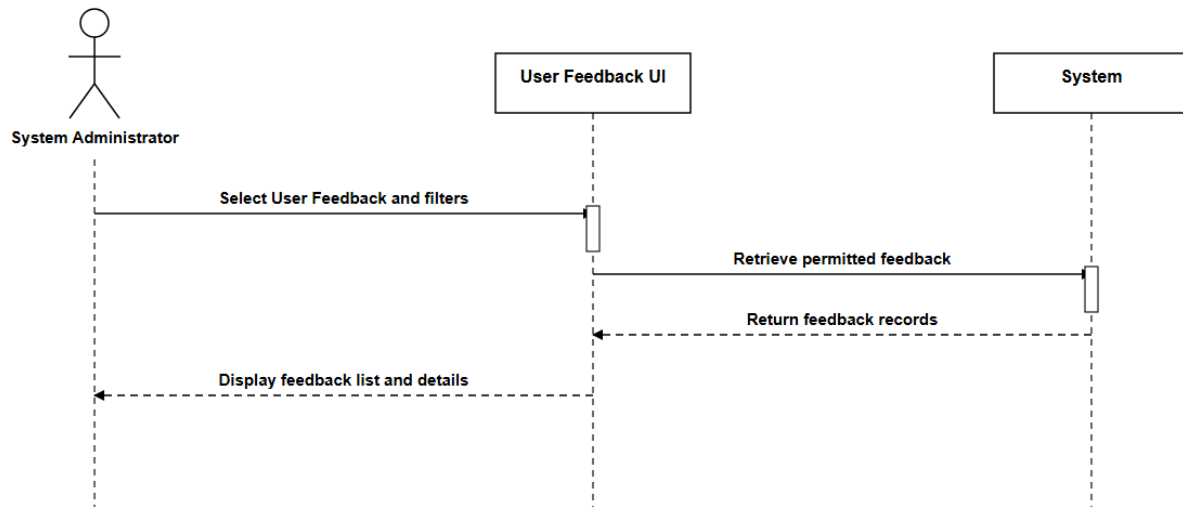
Suspend Account (System Administrator)



Unsuspend Account (System Administrator)

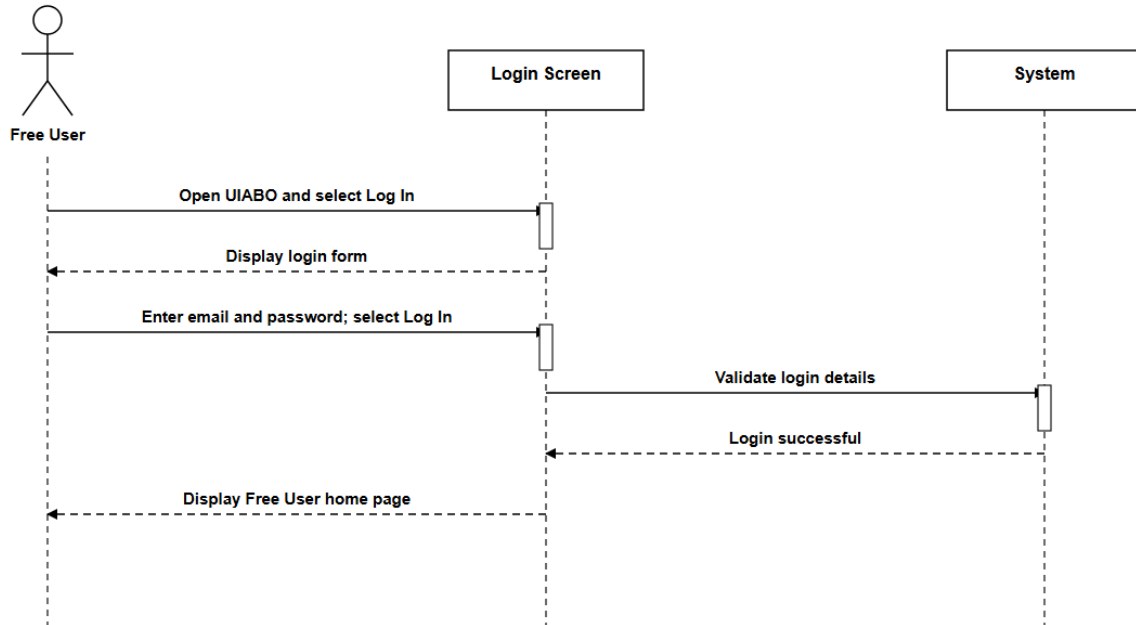


View User Feedback (System Administrator)

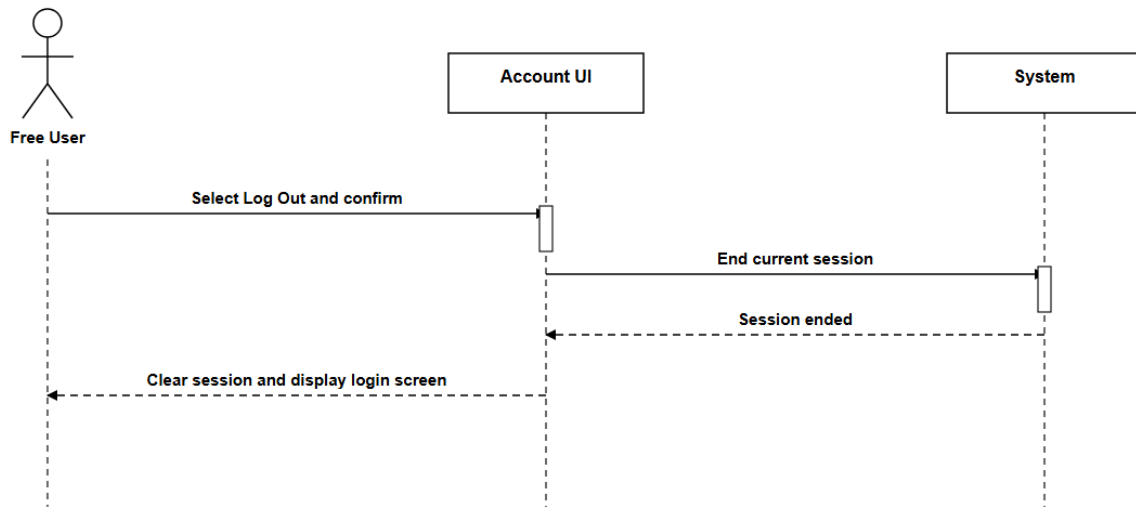


5.2 Free User

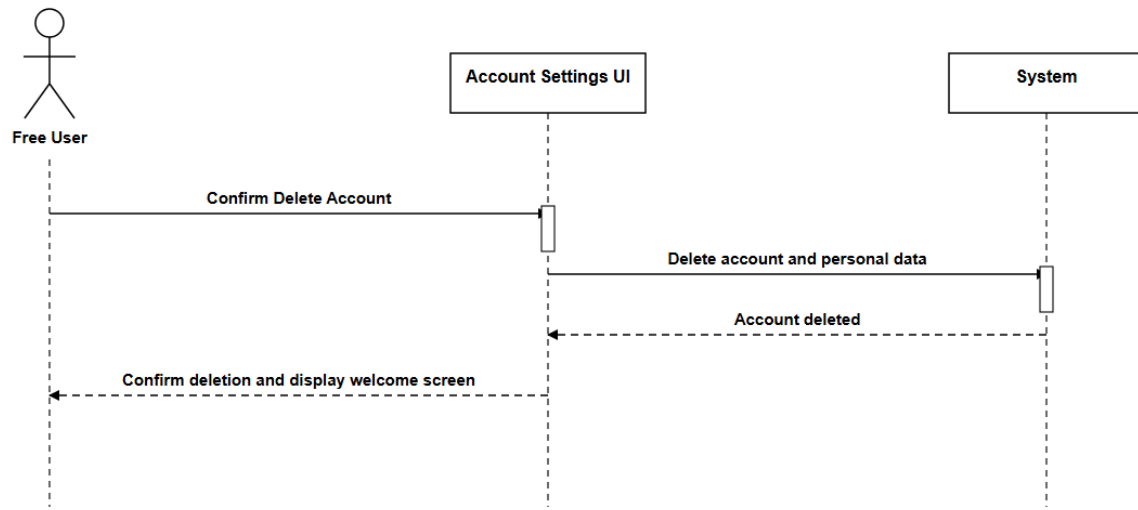
Login (Free User)



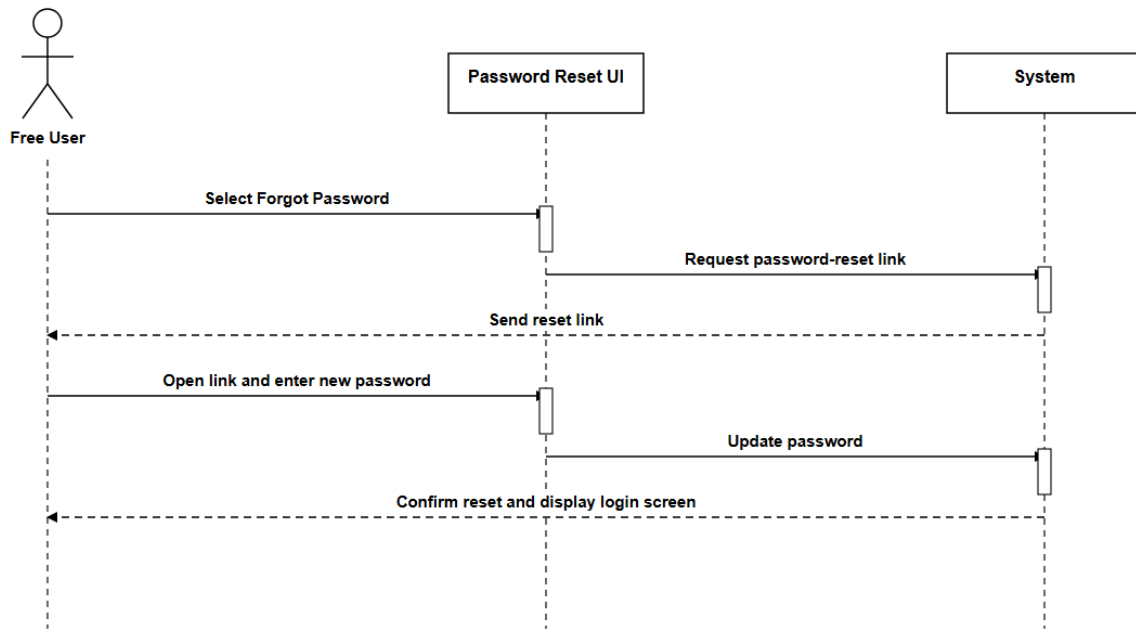
Logout (Free User)



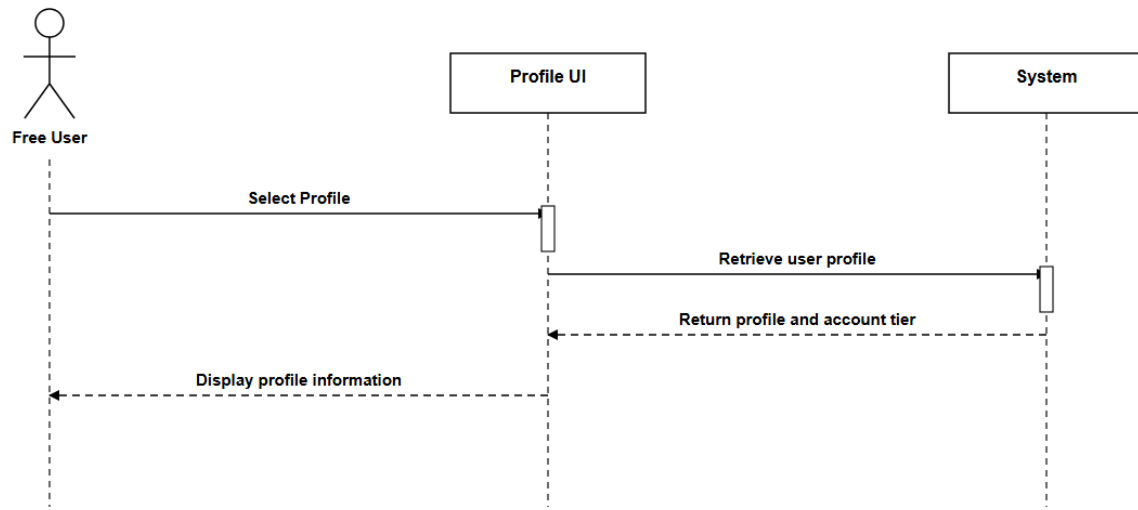
Delete Account (Free User)



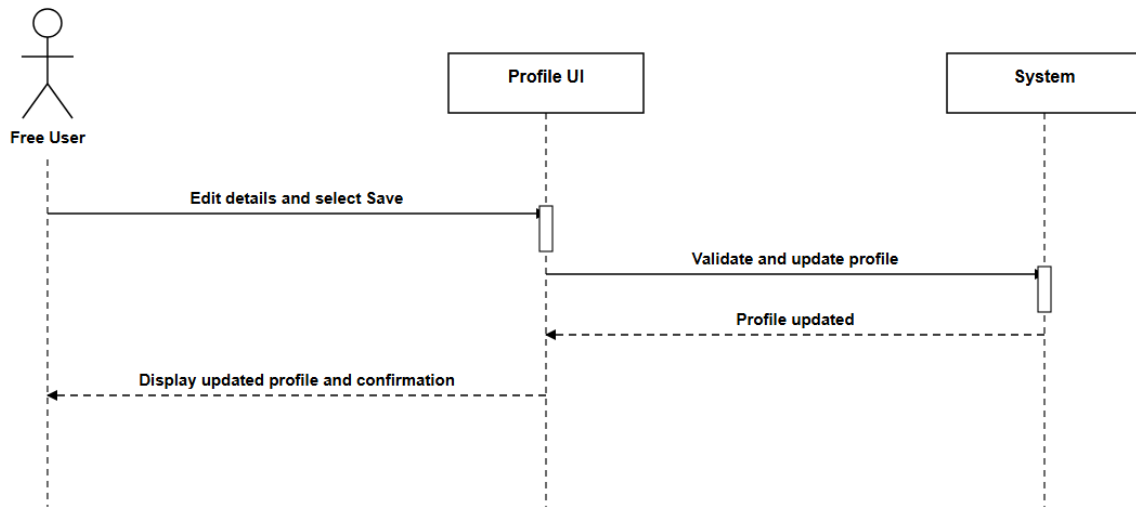
Reset Password (Free User)



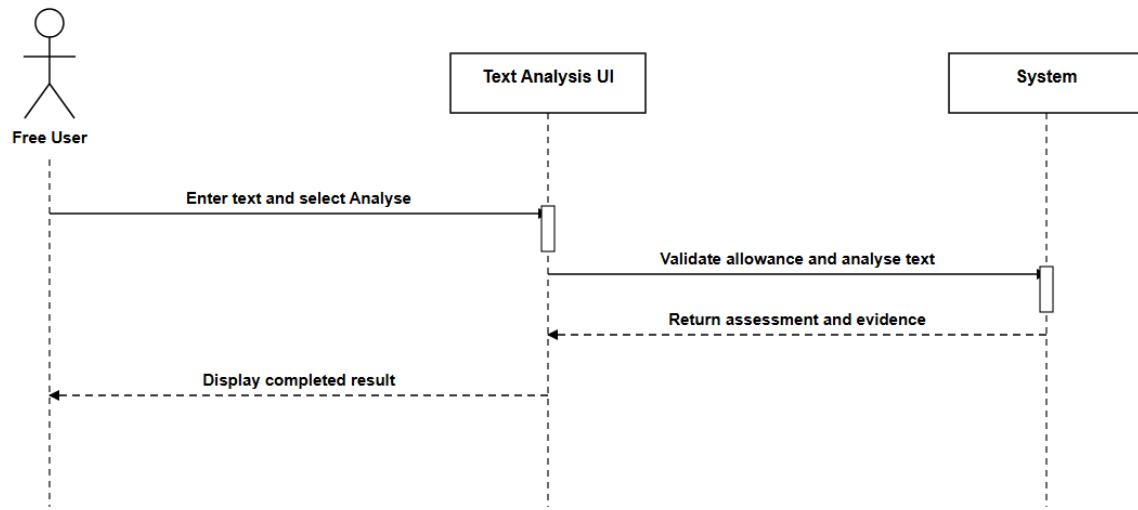
View Profile (Free User)



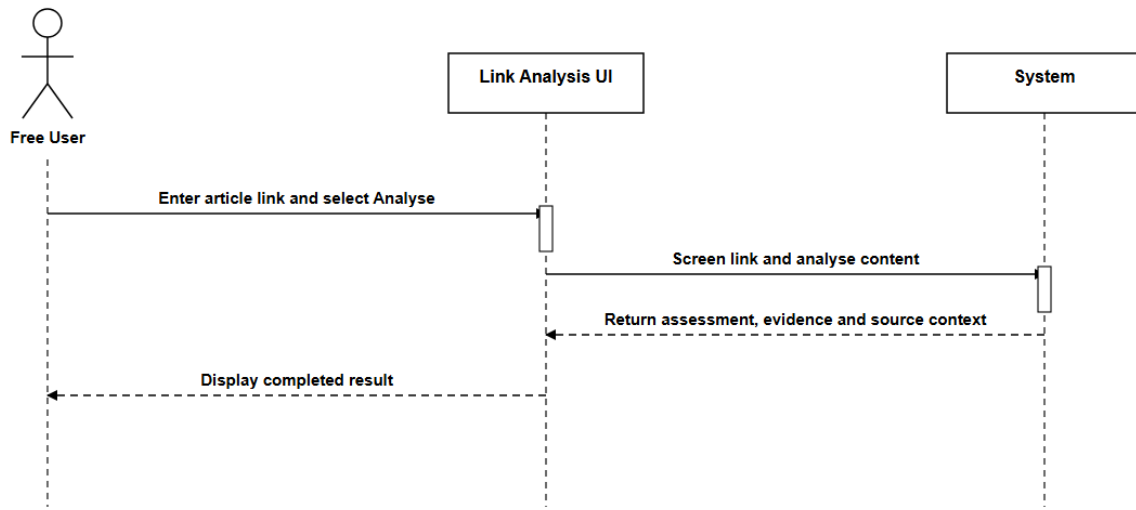
Update Profile (Free User)



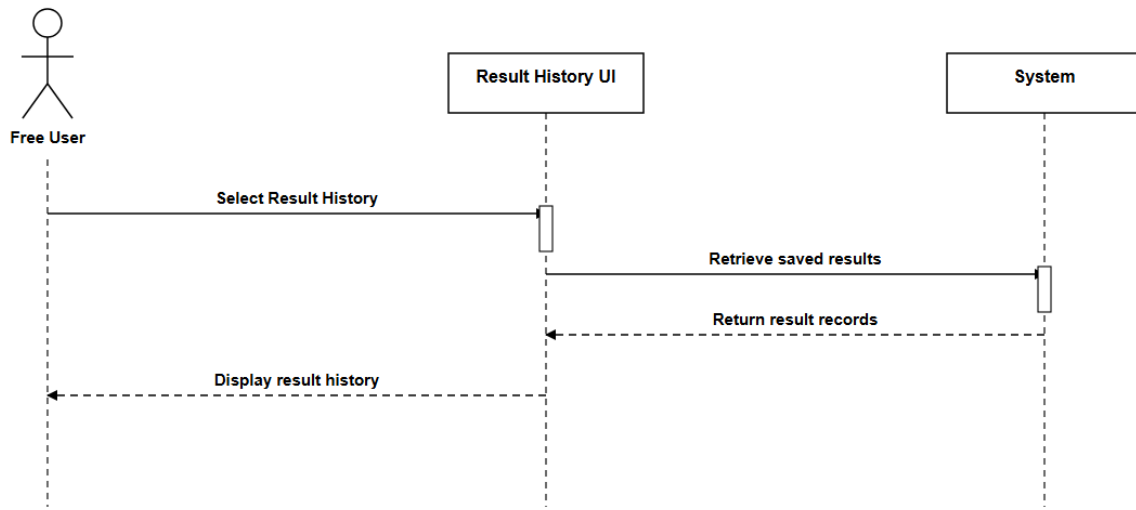
Analyse Short-form Text (Free User)



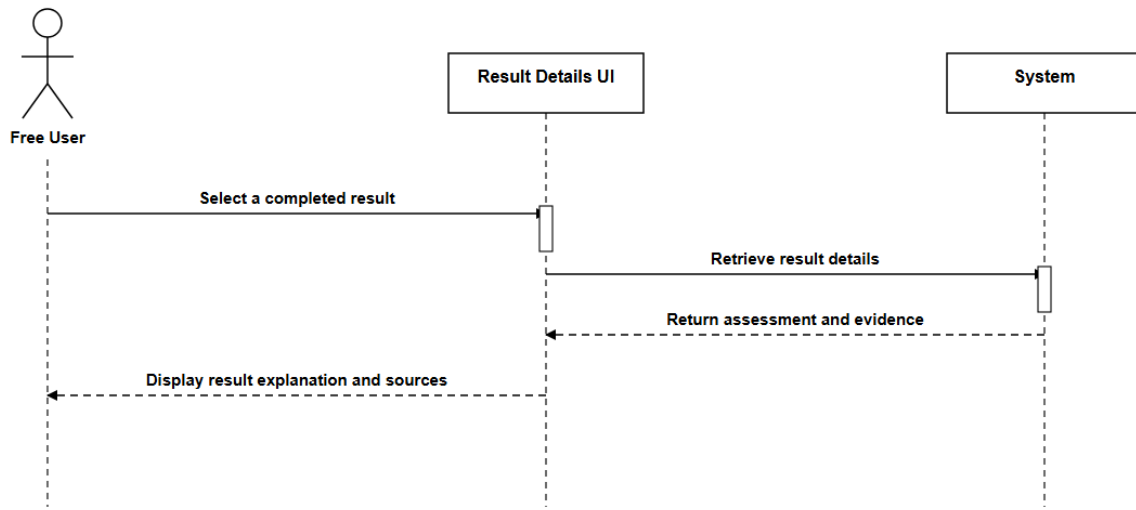
Analyse Public Webpage Link (Free User)



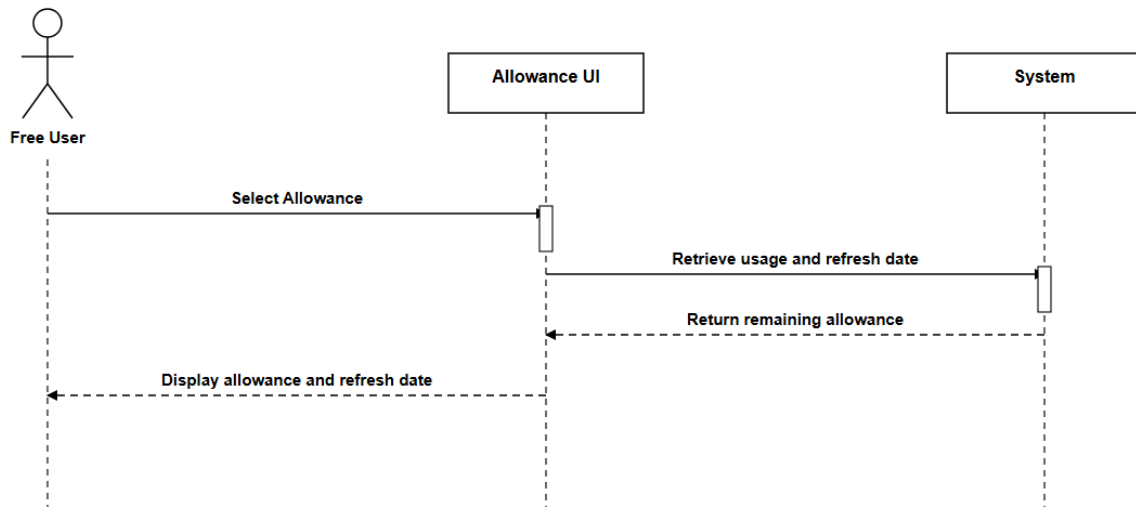
View Result History (Free User)



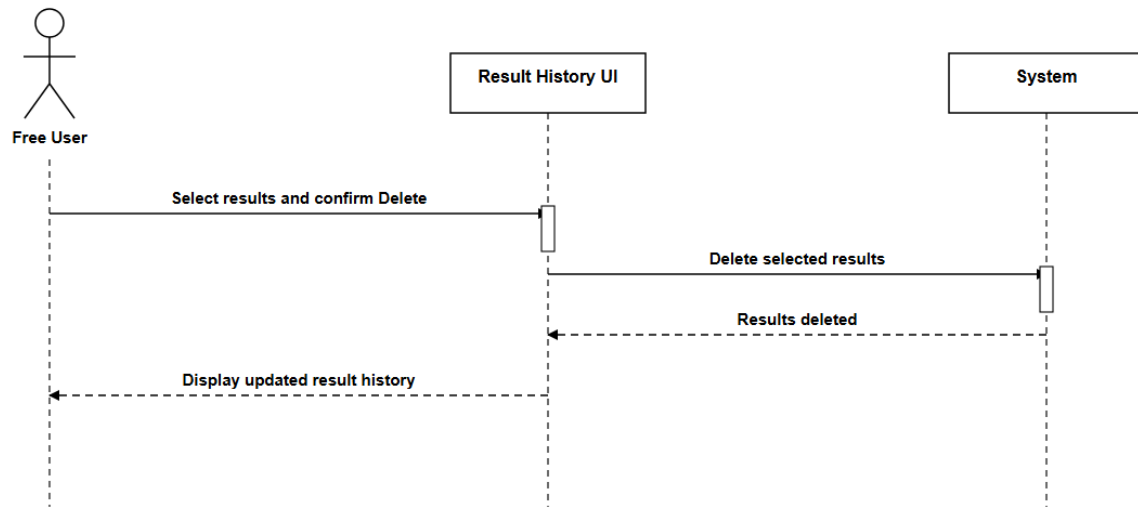
View Result Details (Free User)



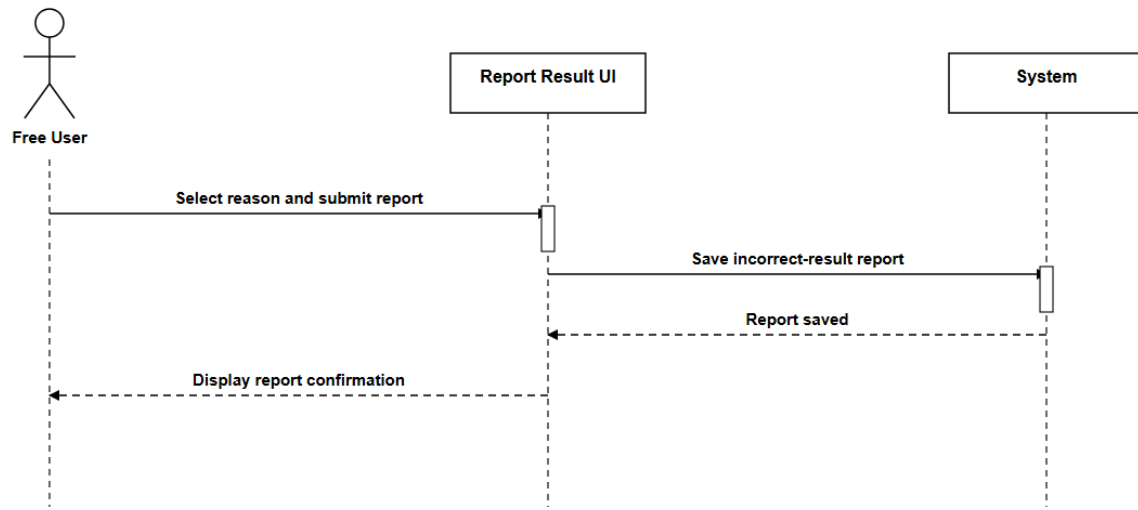
View Remaining Allowance (Free User)



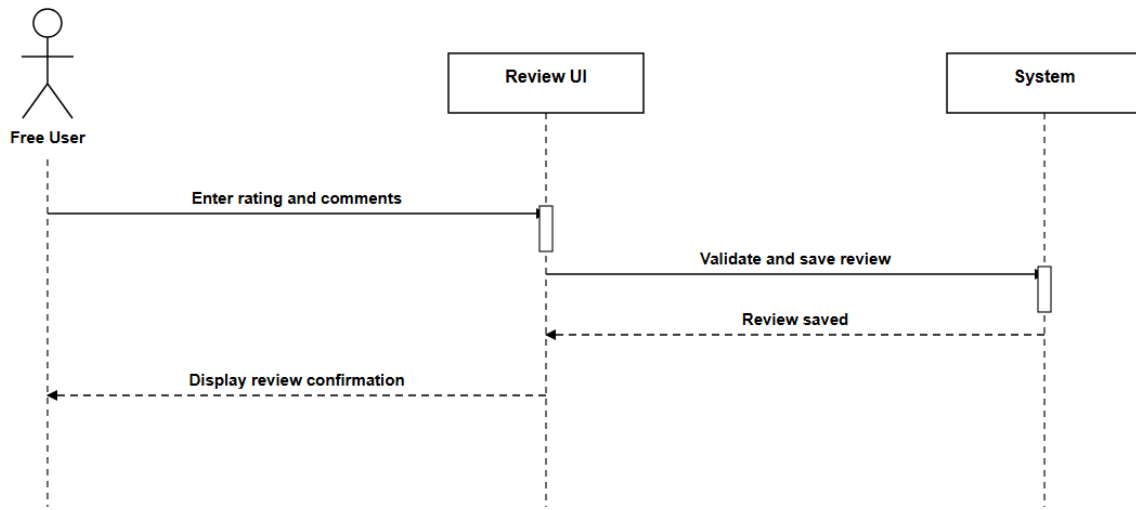
Delete Results (Free User)



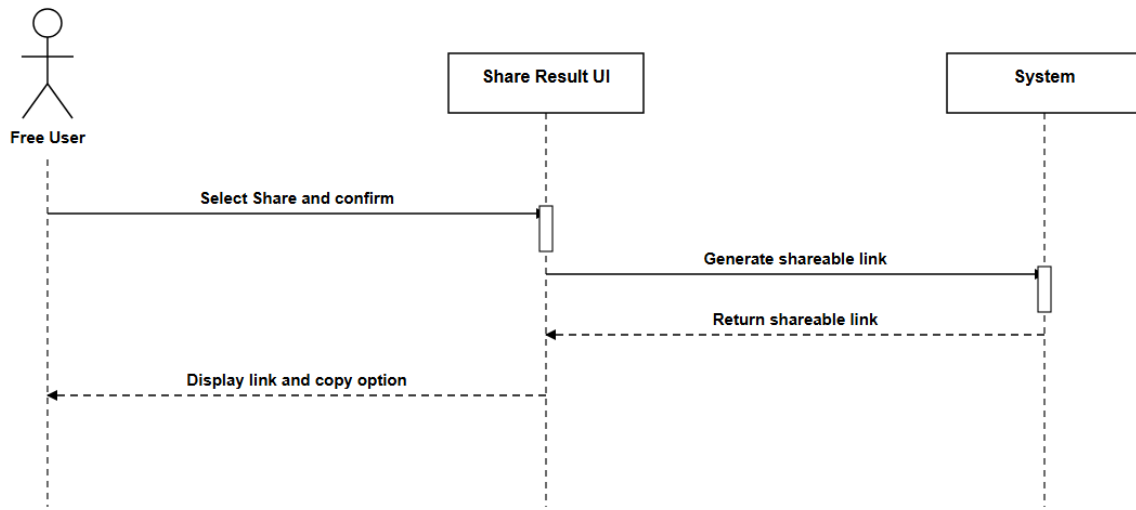
Report Incorrect Result (Free User)



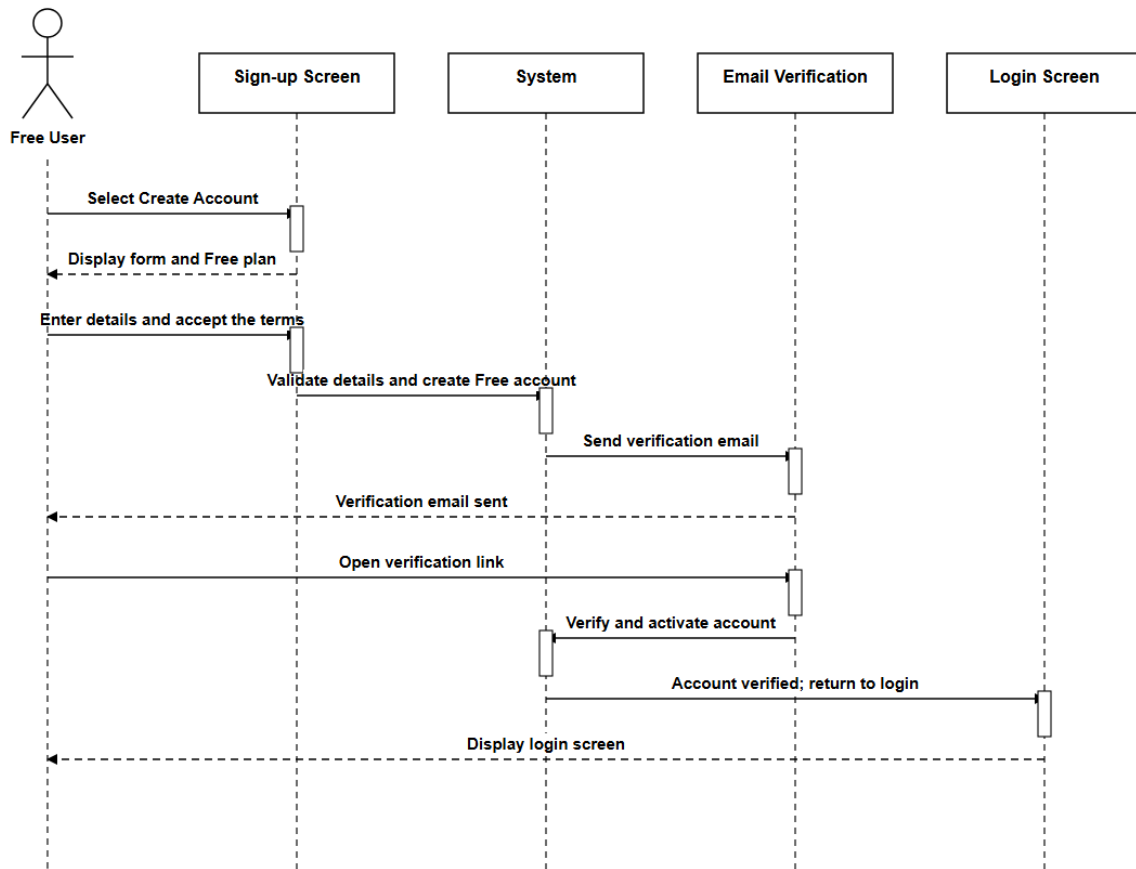
Leave a Review (Free User)



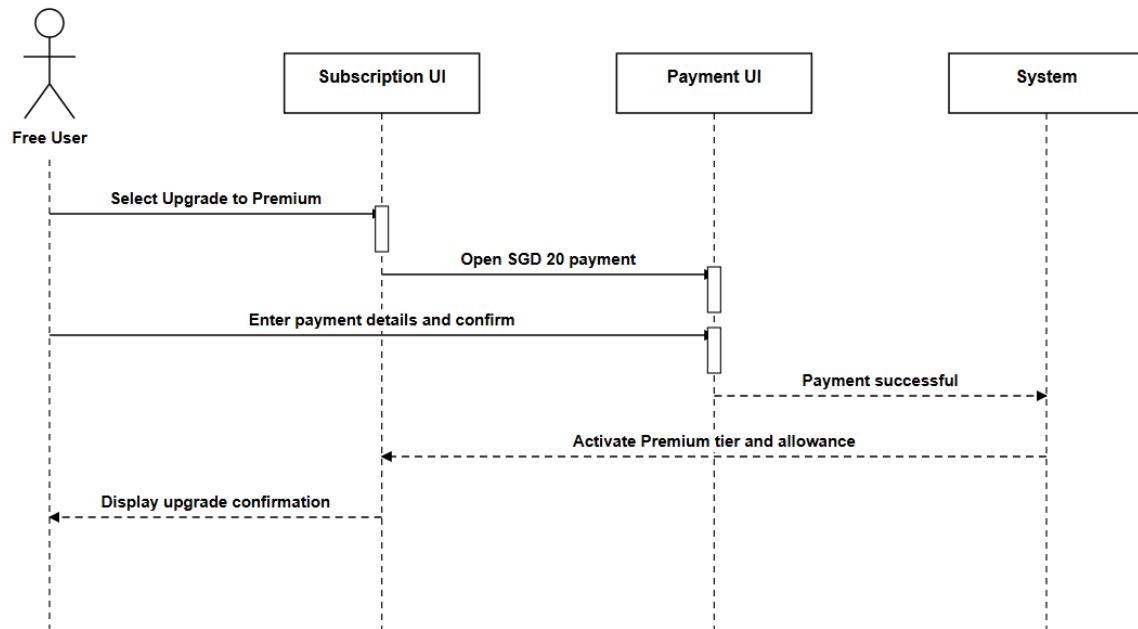
Share Result (Free User)



Create Account (Free User)

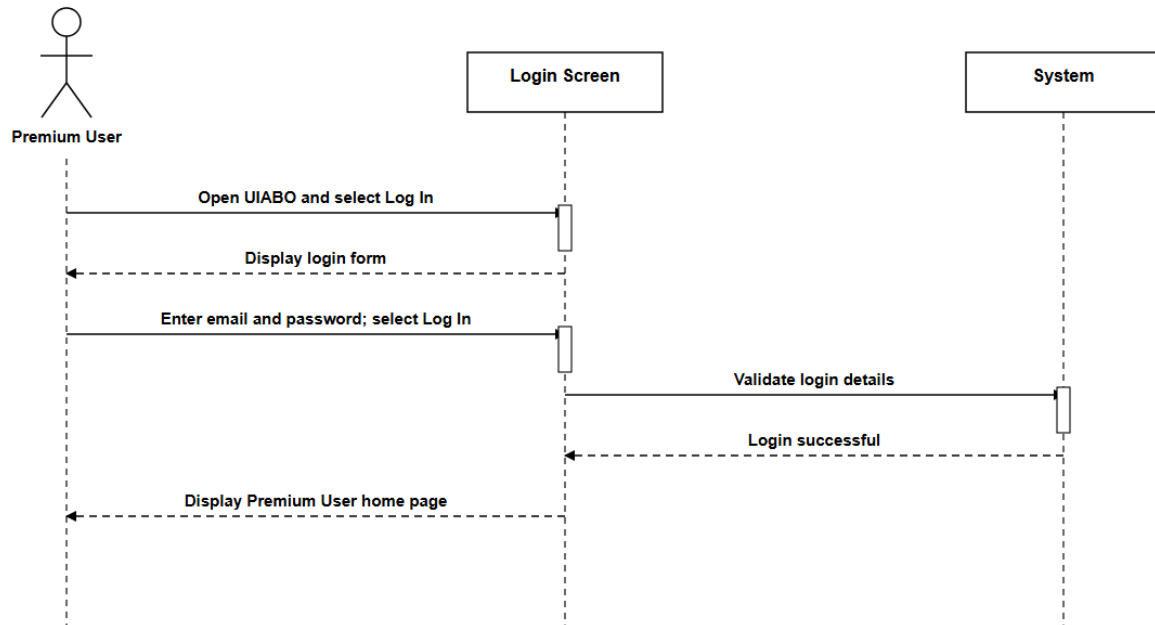


Upgrade to Premium (Free User)

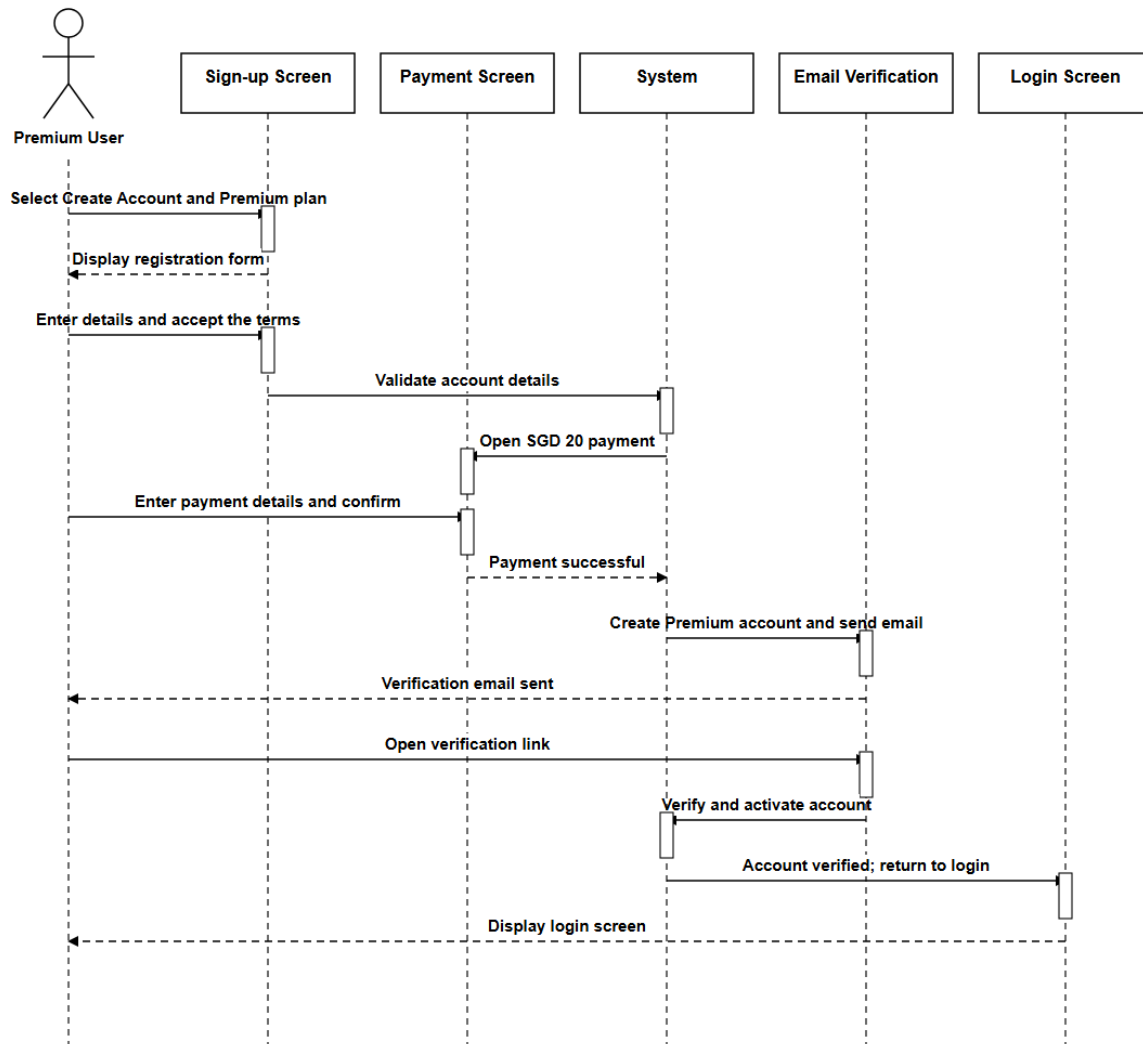


5.3 Premium User

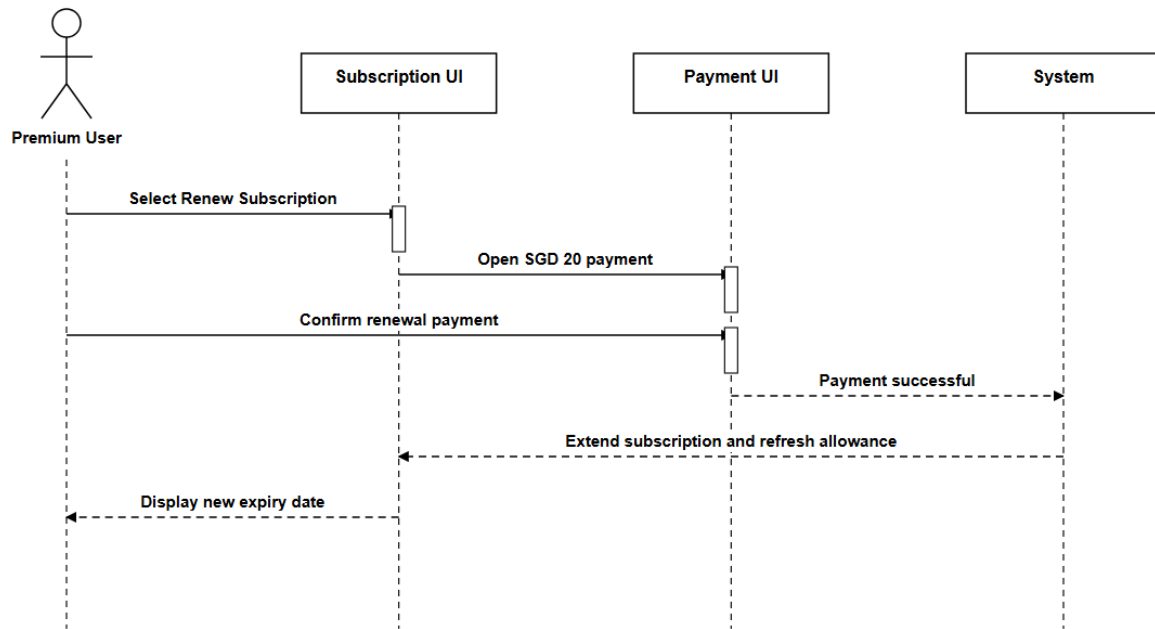
Login (Premium User)



Create Account (Premium User)



Renew Subscription (Premium User)



Cancel Subscription (Premium User)

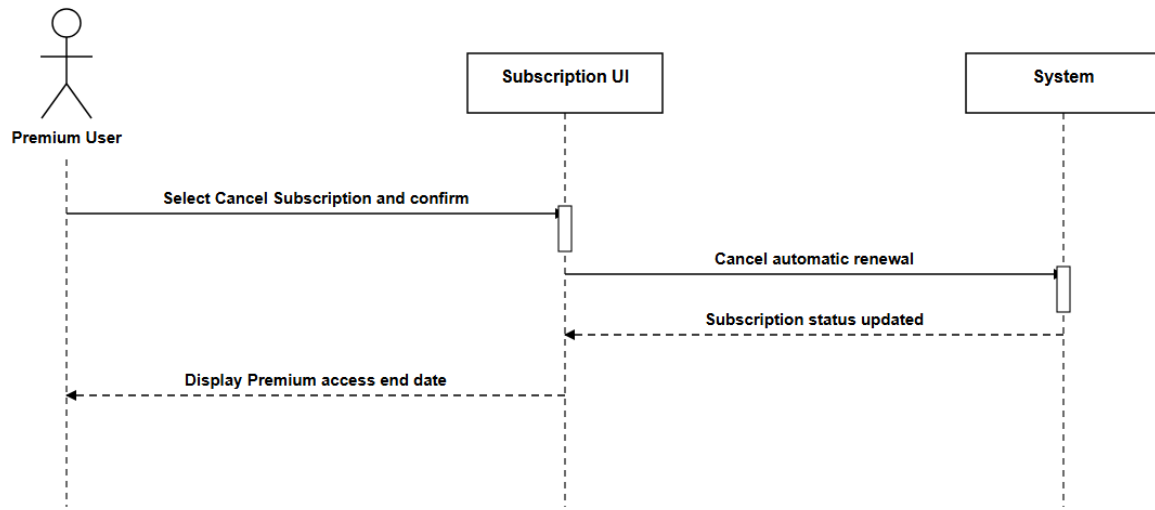
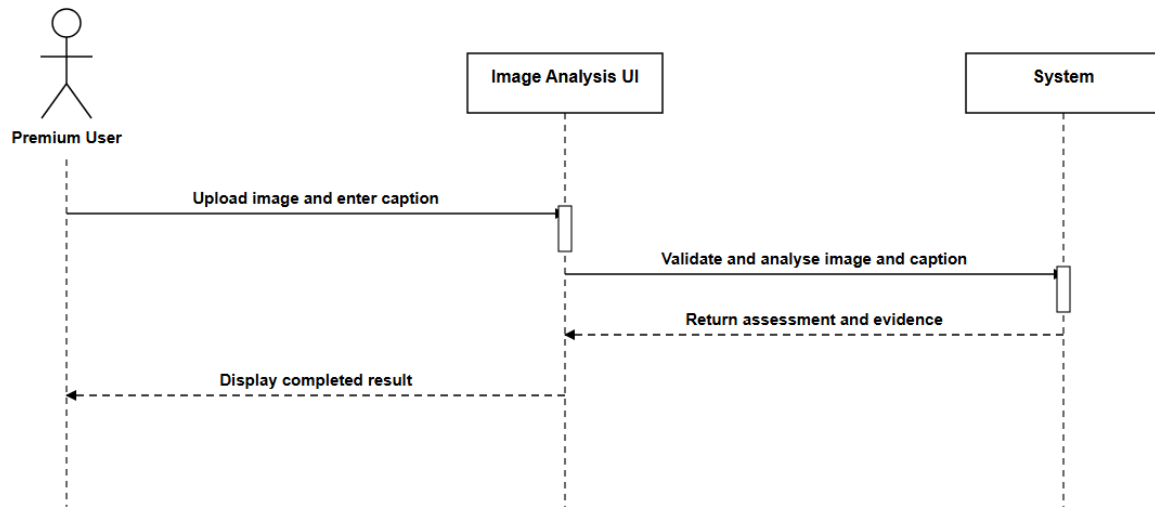
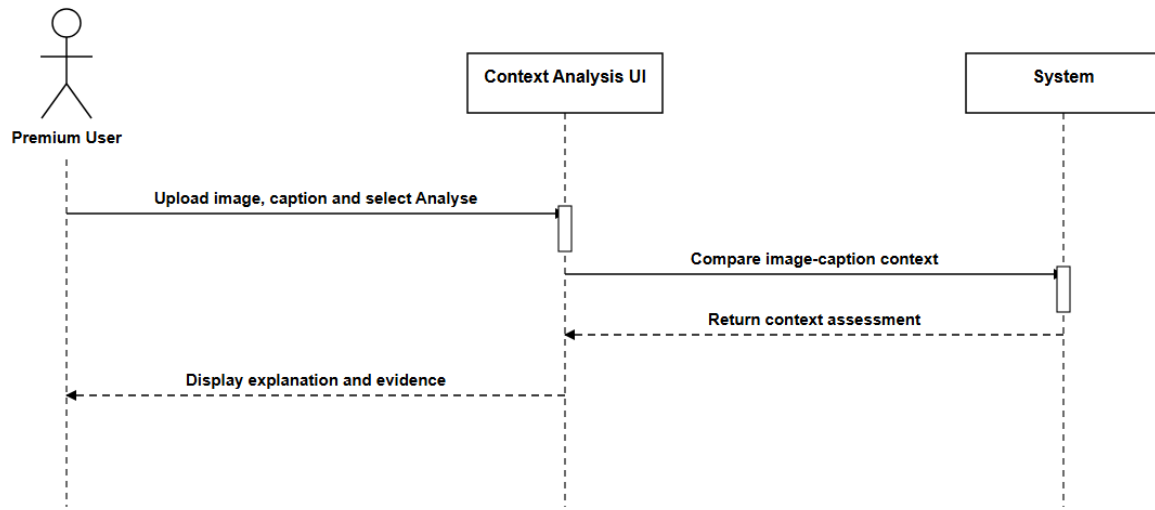


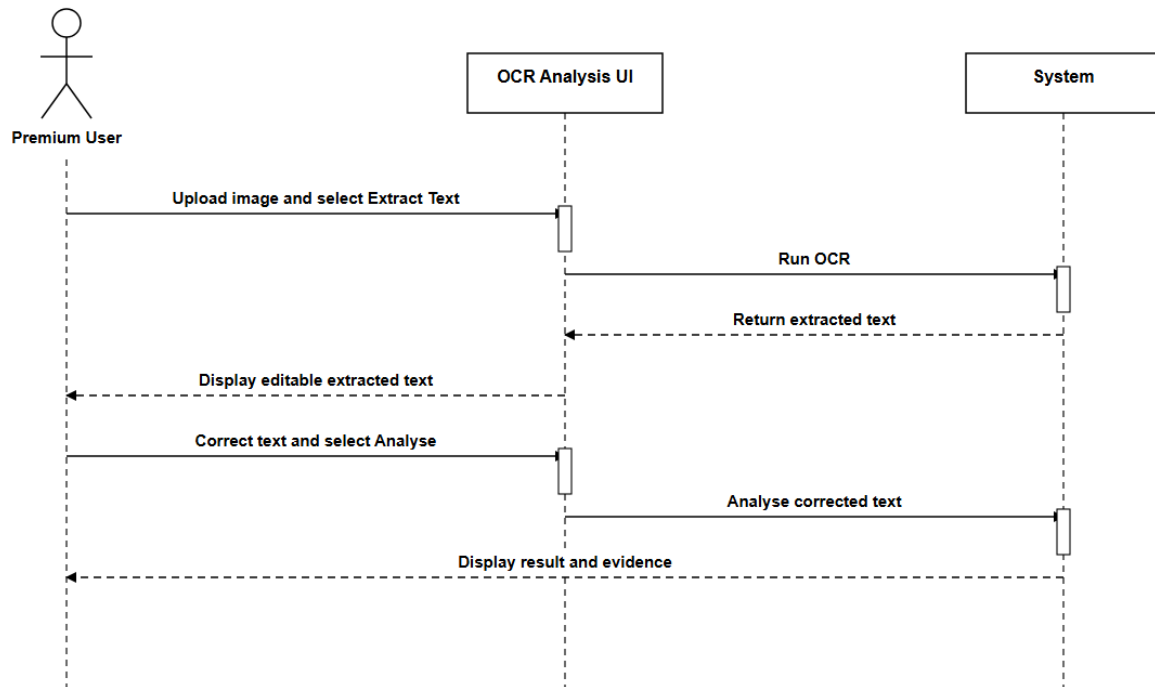
Image and Caption Analysis (Premium User)



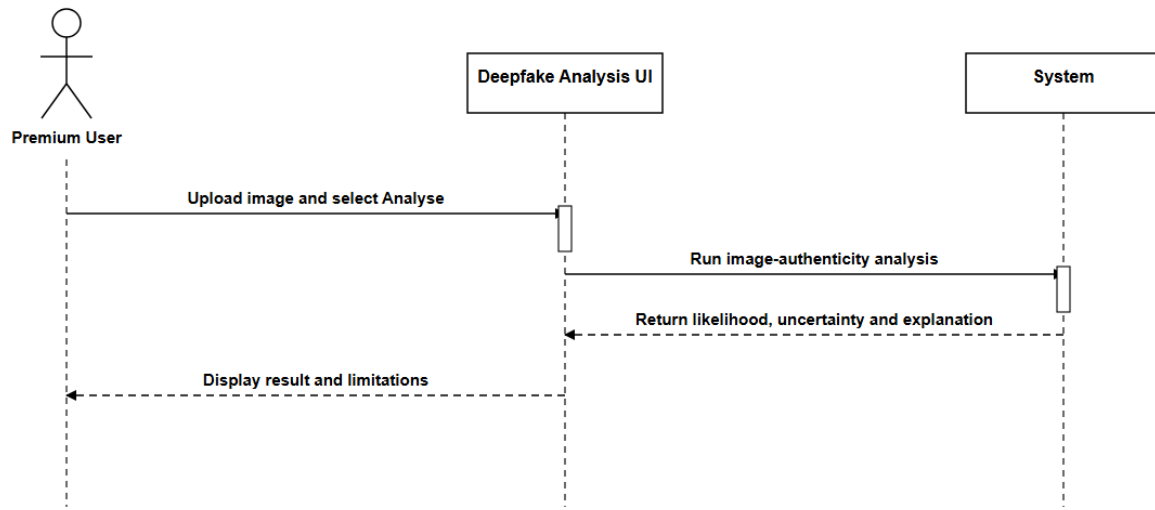
Caption-Image Context Analysis (Premium User)



OCR Image Analysis (Premium User)

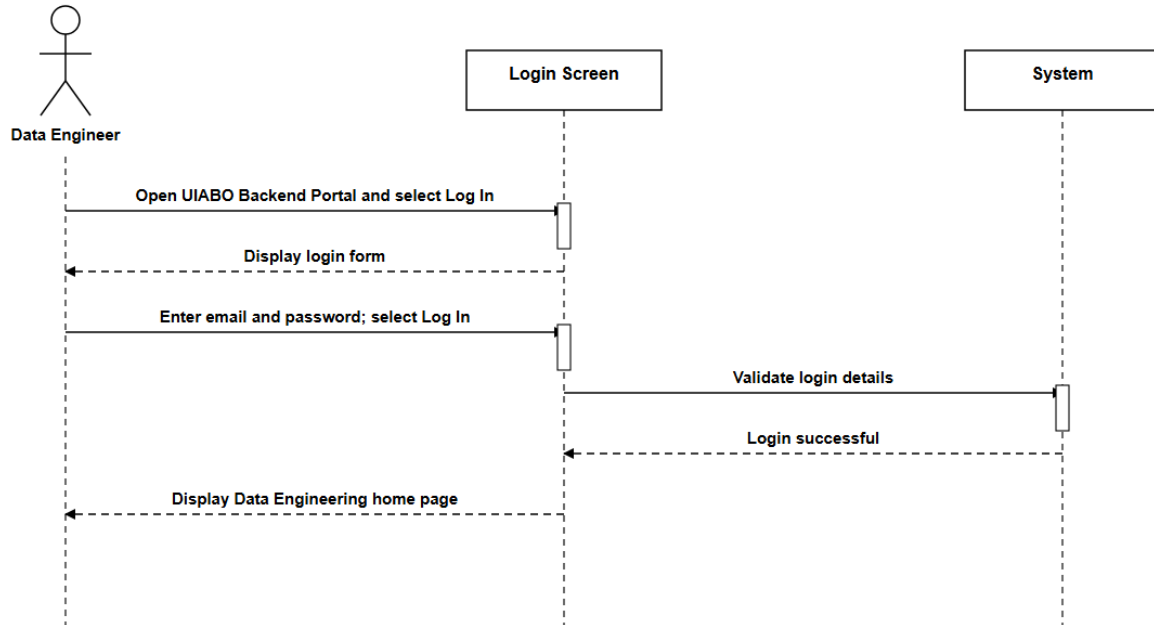


Deepfake/AI Image Analysis (Premium User)

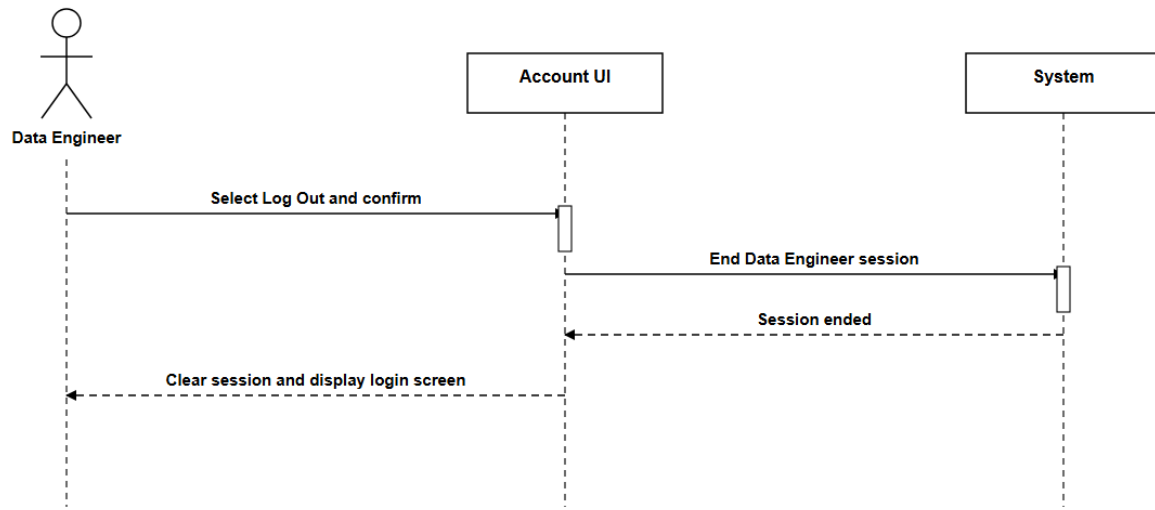


5.4 Data Engineer

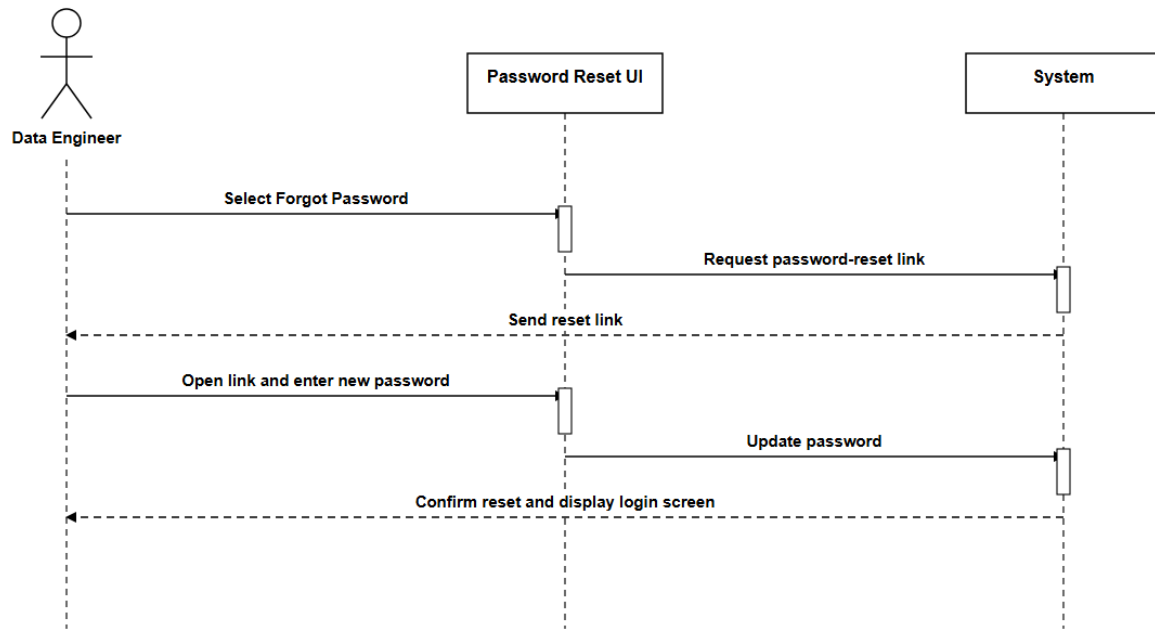
Login (Data Engineer)



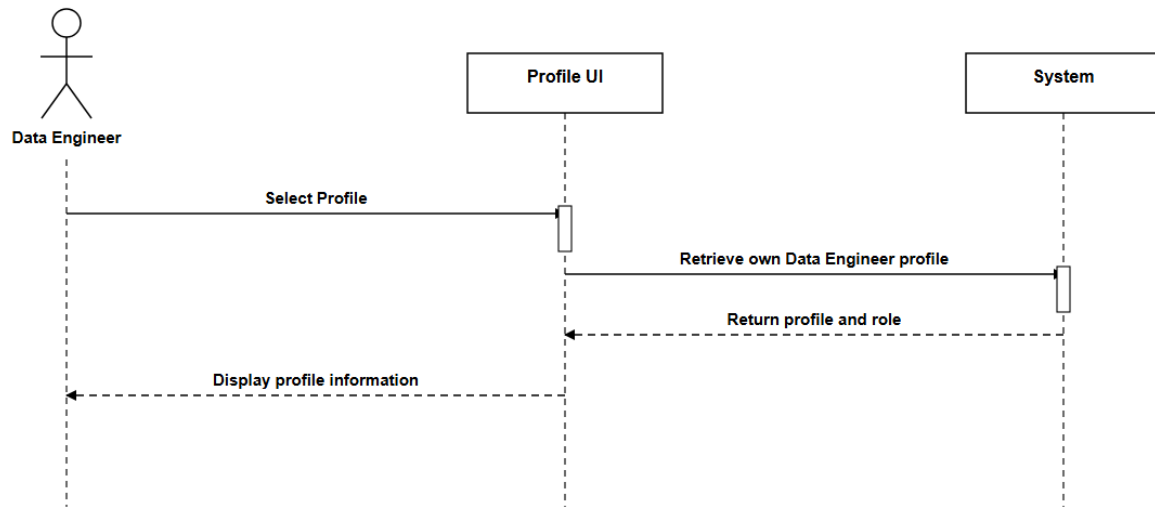
Logout (Data Engineer)



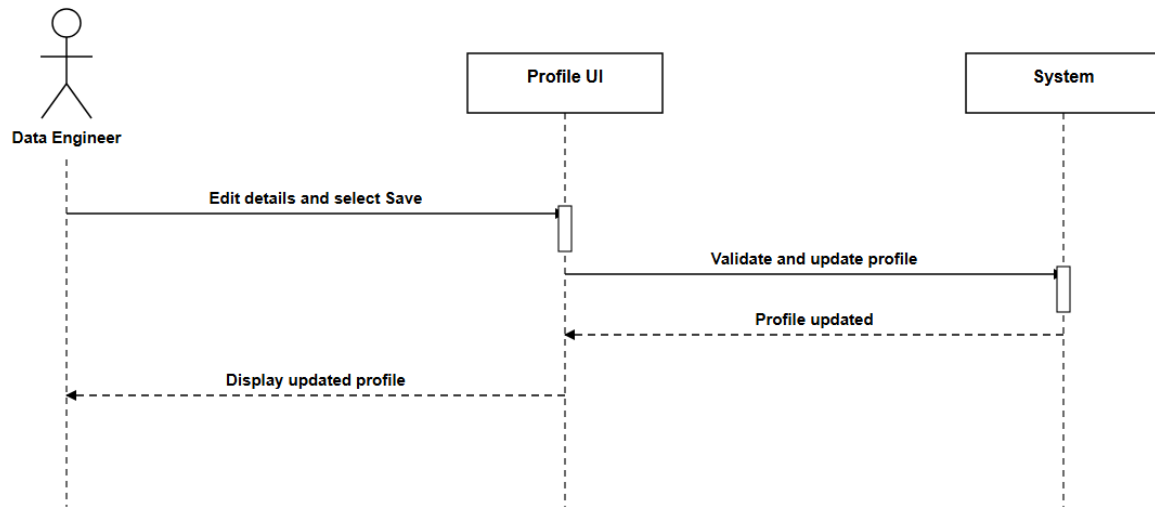
Reset Password (Data Engineer)



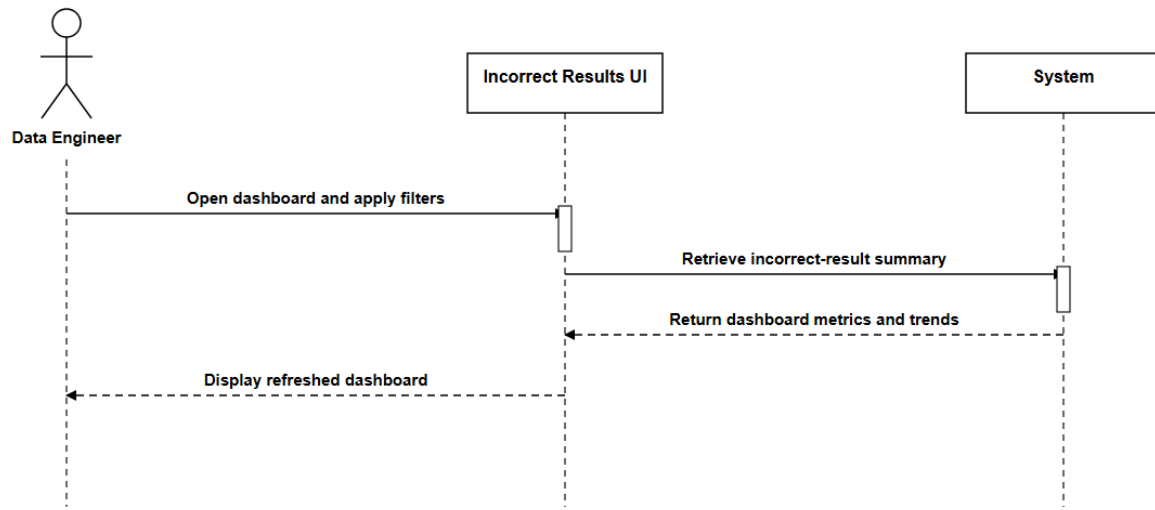
View Profile (Data Engineer)



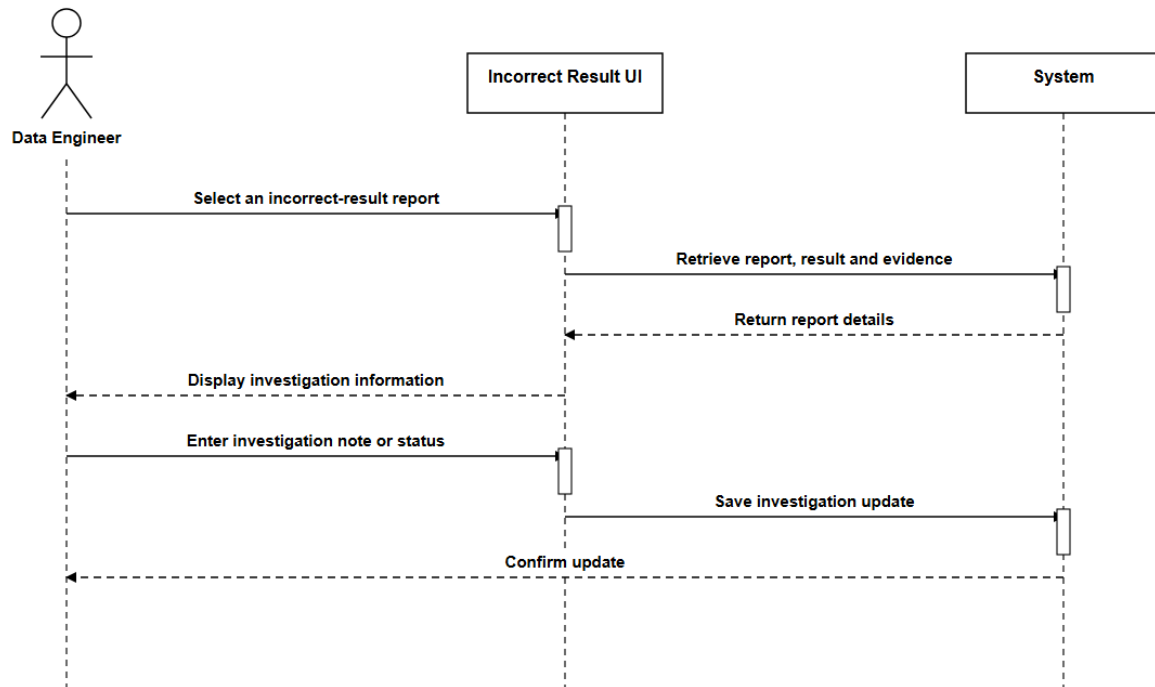
Update Profile (Data Engineer)



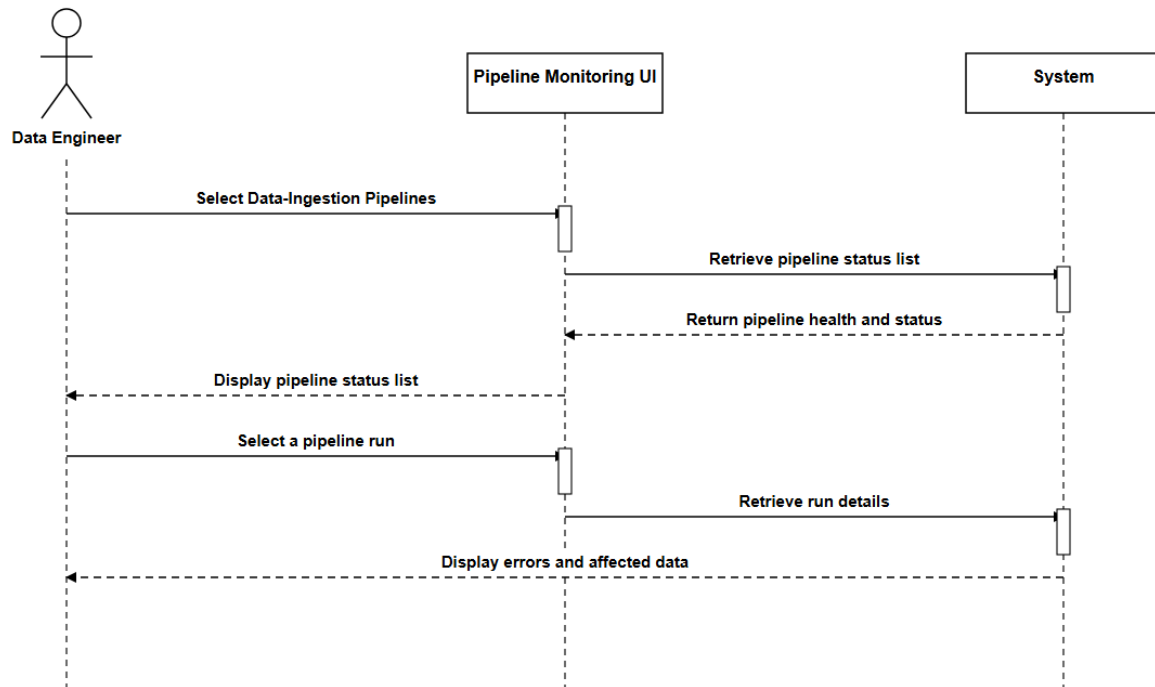
View Incorrect Results Dashboard (Data Engineer)



View Incorrect Result Report (Data Engineer)



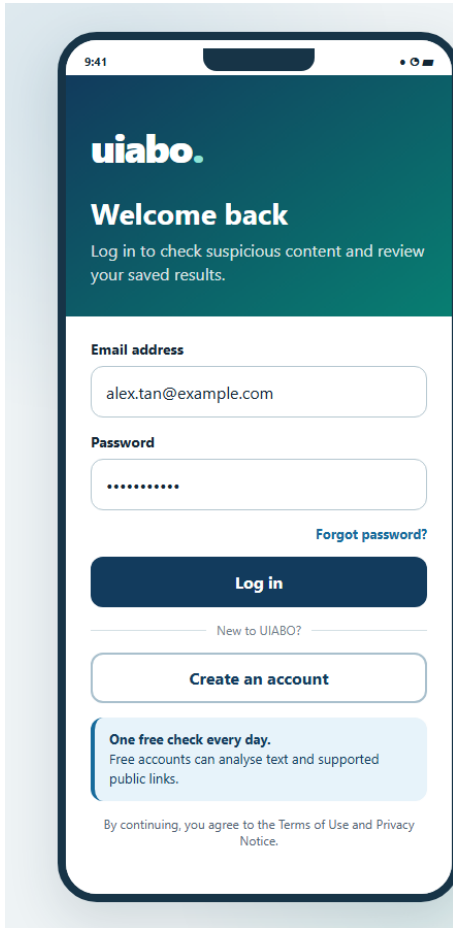
Monitor Data-Ingestion Pipelines (Data Engineer)



6. User Interface

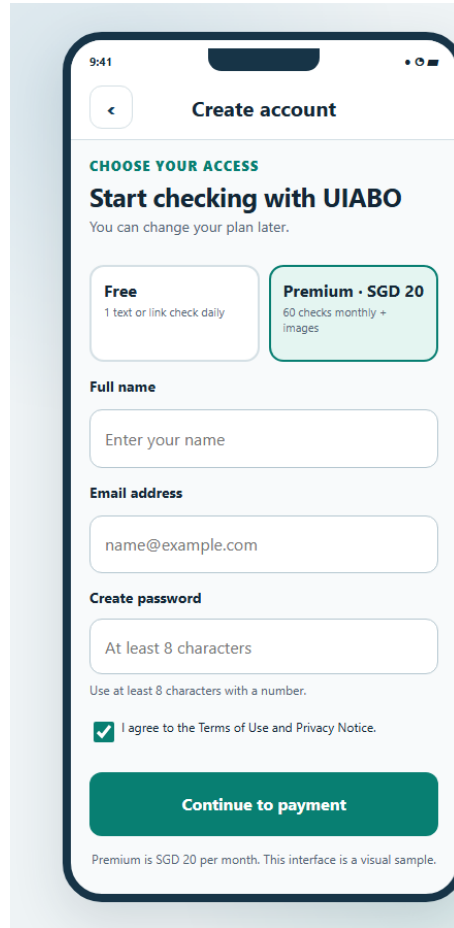
6.1 Login and Account Creation

Login Page



The login screen features a dark teal header with the 'uiabo.' logo and a 'Welcome back' message. Below this, there are input fields for 'Email address' (containing 'alex.tan@example.com') and 'Password' (masked with dots). A 'Forgot password?' link is positioned to the right of the password field. A large dark teal 'Log in' button is centered below the inputs. A link for 'New to UIABO?' leads to the 'Create an account' button. A light blue box highlights the benefit of a free account: 'One free check every day. Free accounts can analyse text and supported public links.' At the bottom, a small disclaimer states: 'By continuing, you agree to the Terms of Use and Privacy Notice.'

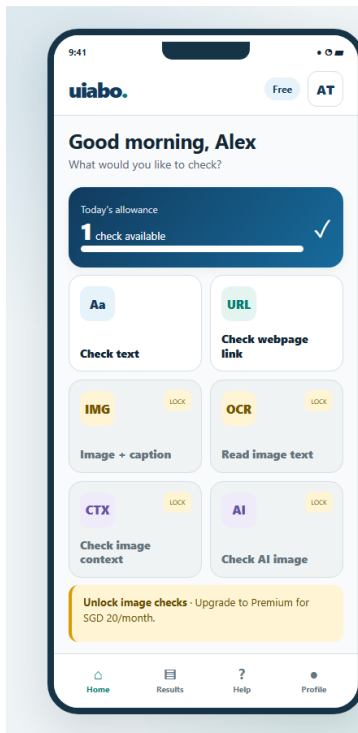
Sign Up Page



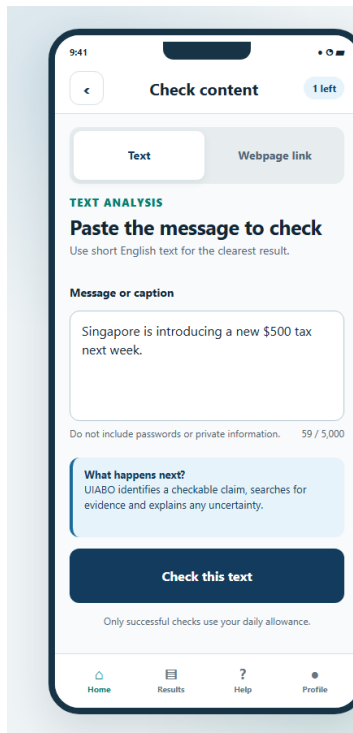
The sign-up screen has a dark teal header with a back arrow and the text 'Create account'. The main heading is 'CHOOSE YOUR ACCESS' followed by 'Start checking with UIABO' and a subtext 'You can change your plan later.' Two plan options are shown: 'Free' (1 text or link check daily) and 'Premium · SGD 20' (60 checks monthly + images). Below these are input fields for 'Full name' and 'Email address' (containing 'name@example.com'). A 'Create password' section includes a field with the placeholder 'At least 8 characters' and a note 'Use at least 8 characters with a number.' A checked checkbox indicates agreement to the 'Terms of Use and Privacy Notice.' A large dark teal 'Continue to payment' button is at the bottom. A small disclaimer at the very bottom reads: 'Premium is SGD 20 per month. This interface is a visual sample.'

6.2 Free User

Home Page
(Premium features locked)



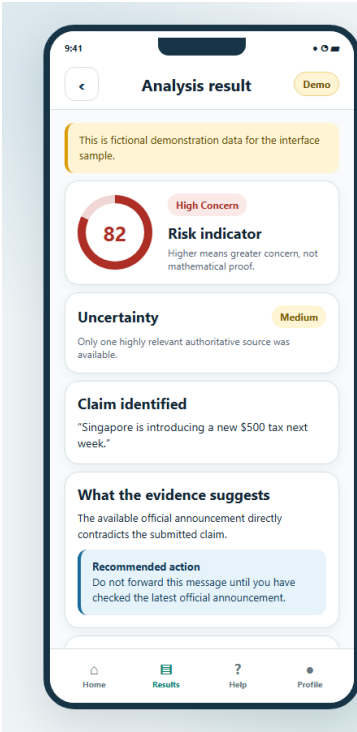
Text Analysis



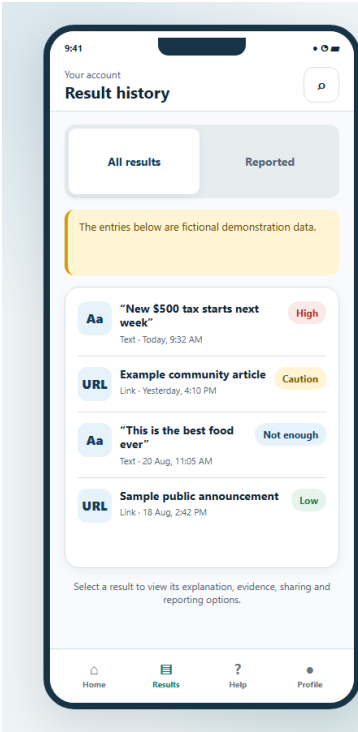
Link Analysis



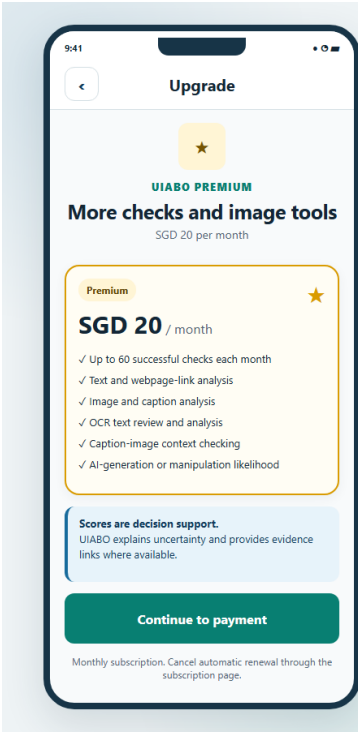
Result Page



Result History



Upgrade to Premium



6.3 Premium User

Home Page

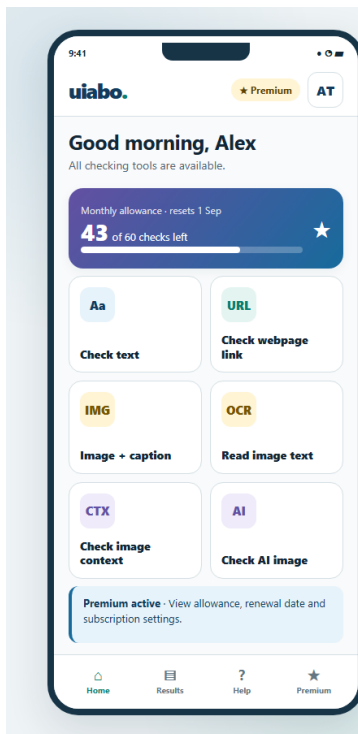
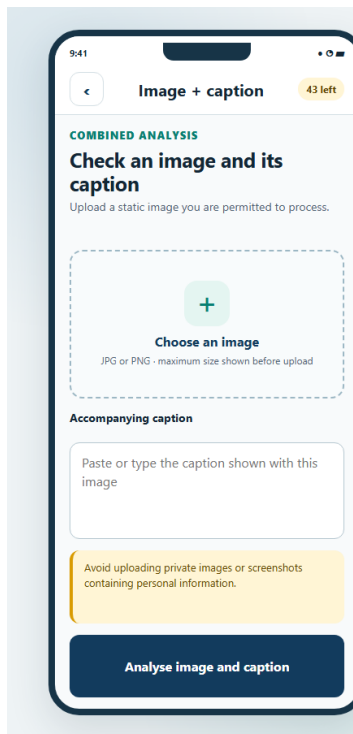
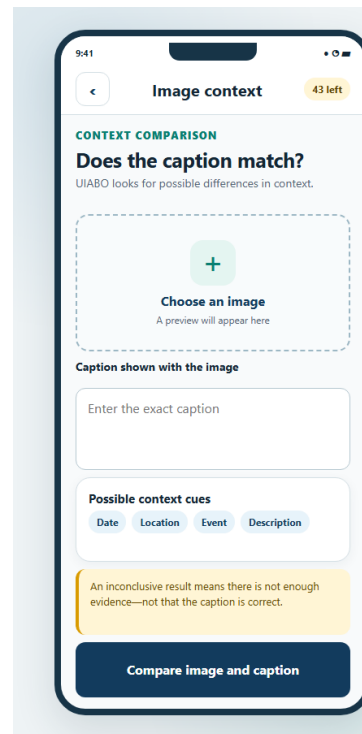


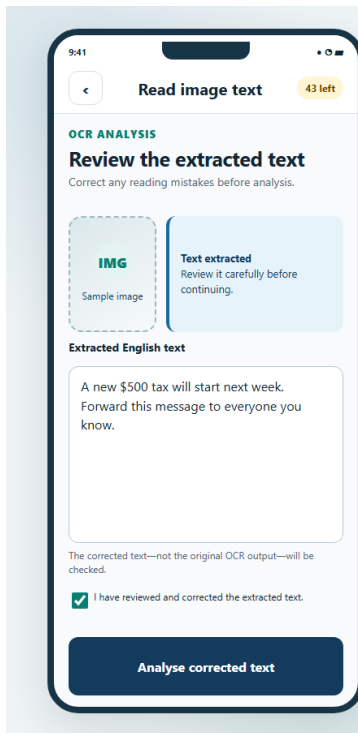
Image Caption Analysis



Context Analysis



OCR Analysis



9:41

< Read image text 43 left

OCR ANALYSIS

Review the extracted text
Correct any reading mistakes before analysis.

IMG
Sample image

Text extracted
Review it carefully before continuing.

Extracted English text

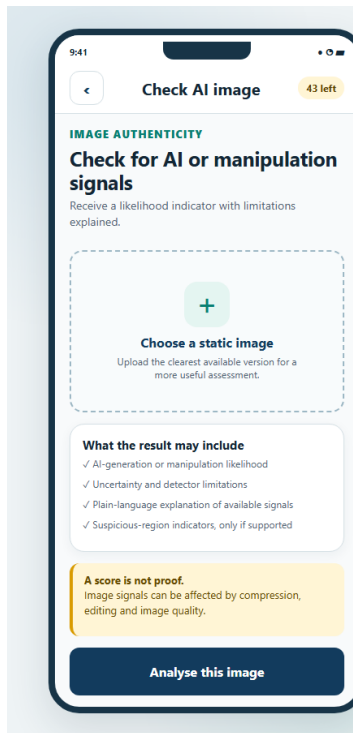
A new \$500 tax will start next week.
Forward this message to everyone you know.

The corrected text—not the original OCR output—will be checked.

☒ I have reviewed and corrected the extracted text.

Analyse corrected text

Deepfake Analysis



9:41

< Check AI image 43 left

IMAGE AUTHENTICITY

Check for AI or manipulation signals
Receive a likelihood indicator with limitations explained.

+

Choose a static image
Upload the clearest available version for a more useful assessment.

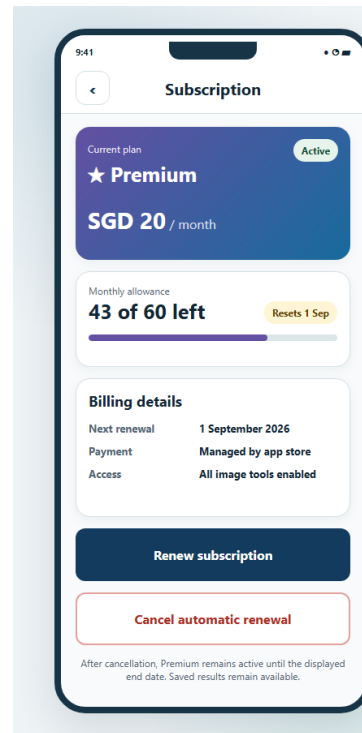
What the result may include

- ✓ AI-generation or manipulation likelihood
- ✓ Uncertainty and detector limitations
- ✓ Plain-language explanation of available signals
- ✓ Suspicious-region indicators, only if supported

A score is not proof.
Image signals can be affected by compression, editing and image quality.

Analyse this image

Subscription



9:41

< Subscription

Current plan **★ Premium** **Active**

SGD 20 / month

Monthly allowance
43 of 60 left **Resets 1 Sep**

Billing details

Next renewal	1 September 2026
Payment	Managed by app store
Access	All image tools enabled

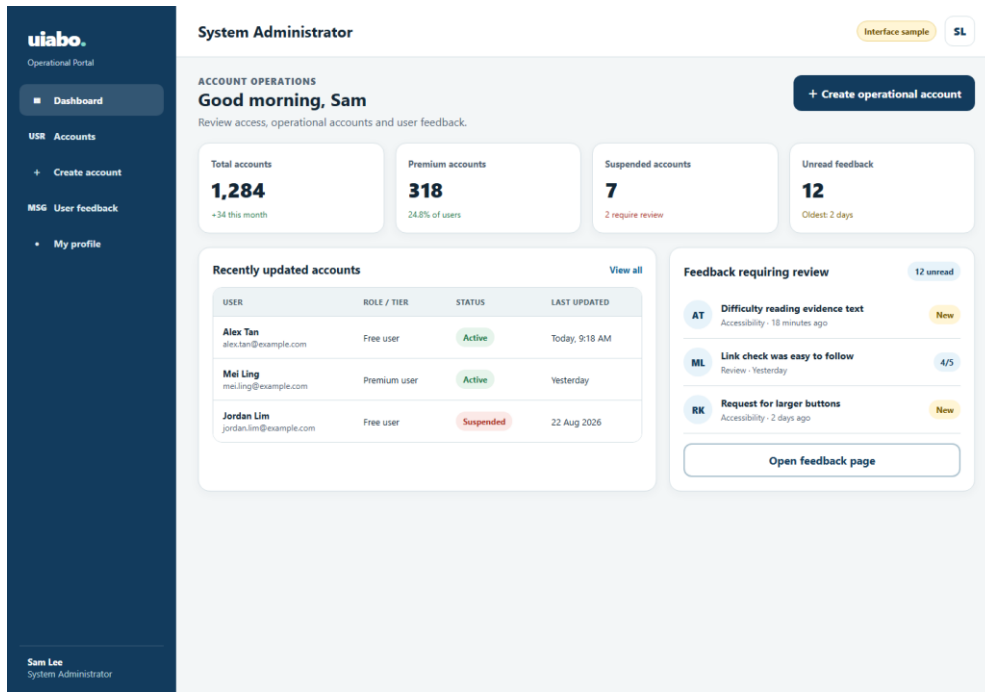
Renew subscription

Cancel automatic renewal

After cancellation, Premium remains active until the displayed end date. Saved results remain available.

6.4 System Administrator

System Administrator Dashboard



The System Administrator Dashboard for uiabo. features a dark blue sidebar with navigation links: Dashboard, Accounts, Create account, User feedback, and My profile. The main content area is titled 'System Administrator' and includes a greeting 'Good morning, Sam'. It displays four summary cards: Total accounts (1,284), Premium accounts (318), Suspended accounts (7), and Unread feedback (12). Below these are two sections: 'Recently updated accounts' with a table of user updates and 'Feedback requiring review' with a list of items needing attention. A '+ Create operational account' button is located in the top right.

System Administrator

ACCOUNT OPERATIONS

Good morning, Sam

Review access, operational accounts and user feedback.

Total accounts
1,284
+34 this month

Premium accounts
318
24.8% of users

Suspended accounts
7
2 require review

Unread feedback
12
Oldest: 2 days

Recently updated accounts

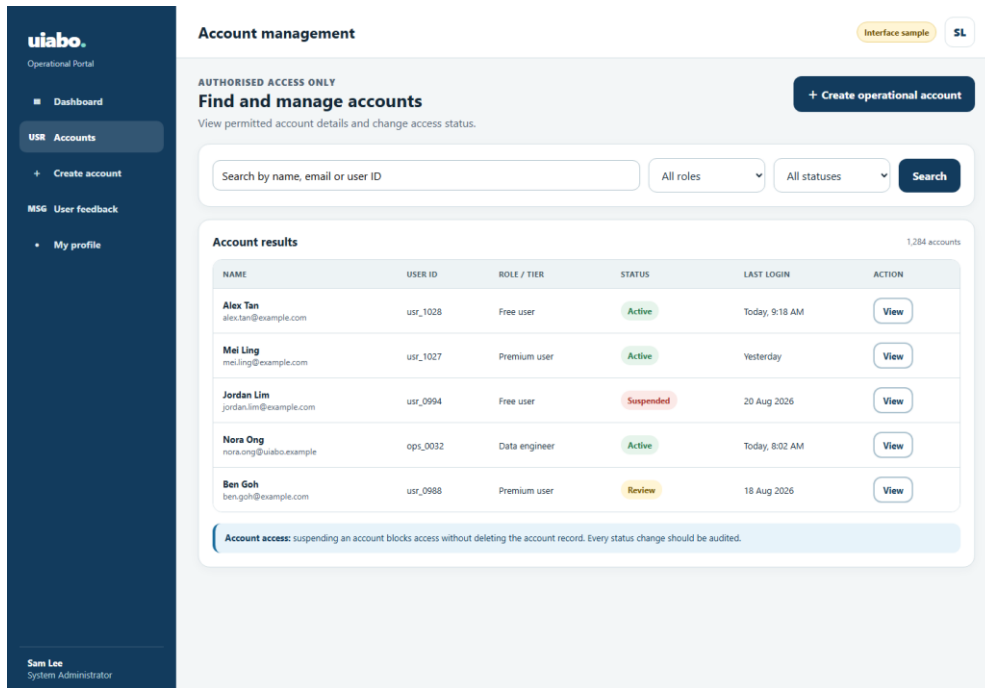
USER	ROLE / TIER	STATUS	LAST UPDATED
Alex Tan alex.tan@example.com	Free user	Active	Today, 9:18 AM
Mei Ling mei.ling@example.com	Premium user	Active	Yesterday
Jordan Lim jordan.lim@example.com	Free user	Suspended	22 Aug 2026

Feedback requiring review

- AT Difficulty reading evidence text
Accessibility - 18 minutes ago
- ML Link check was easy to follow
Review - Yesterday
- RK Request for larger buttons
Accessibility - 2 days ago

Open feedback page

Account Management



The Account Management Dashboard for uiabo. features a dark blue sidebar with navigation links: Dashboard, Accounts, Create account, User feedback, and My profile. The main content area is titled 'Account management' and includes a greeting 'Find and manage accounts'. It displays a search bar, filters for roles and statuses, and a table of account results. A '+ Create operational account' button is located in the top right.

Account management

AUTHORISED ACCESS ONLY

Find and manage accounts

View permitted account details and change access status.

Search by name, email or user ID

All roles

All statuses

Search

Account results

NAME	USER ID	ROLE / TIER	STATUS	LAST LOGIN	ACTION
Alex Tan alex.tan@example.com	usr_1028	Free user	Active	Today, 9:18 AM	View
Mei Ling mei.ling@example.com	usr_1027	Premium user	Active	Yesterday	View
Jordan Lim jordan.lim@example.com	usr_0994	Free user	Suspended	20 Aug 2026	View
Nora Ong nora.ong@uiabo.example	opt_0032	Data engineer	Active	Today, 8:02 AM	View
Ben Goh ben.goh@example.com	usr_0988	Premium user	Review	18 Aug 2026	View

Account access: suspending an account blocks access without deleting the account record. Every status change should be audited.

Create Operational Account

uiabo.

Operational Portal

Dashboard

Accounts

Create account

User feedback

My profile

Sam Lee
System Administrator

Create operational account

Interface sample SL

RESTRICTED ADMINISTRATOR FUNCTION

New operational user

Create an authorised system-administrator or data-engineer account.

Account information

Approval required

Full name

Enter legal name

Work email

name@uiabo.example

Operational role

Data Engineer

Initial status

Pending activation

Business reason

Explain why this operational access is required

☐ I have verified that this person is authorised to receive operational access.

Cancel

Create and send invite

!

Access is role-specific

System administrators manage accounts and feedback.
Data engineers investigate incorrect results and monitor data-ingestion pipelines.

Security reminder

Use the minimum necessary role. Operational accounts must not be shared.

After creation

1. The user receives an invitation.
2. Email ownership is verified.
3. The account remains auditable.

User Feedback

uiabo.

Operational Portal

Dashboard

Accounts

Create account

User feedback

My profile

Sam Lee
System Administrator

User feedback

Interface sample SL

REVIEWS AND EXPERIENCE

Feedback from UIABO users

Review comments about accessibility, usability and the service experience.

Average review

4.2/5

Based on demo data

Unread

12

Requires review

Accessibility

8

This month

Resolved

37

Last 30 days

Latest feedback

All categories

AT

Some evidence text is difficult to read

Accessibility - "Please make the source summary larger."

New

ML

The link checker was easy to follow

Review - Rated 4 out of 5

Review

RK

Request for clearer locked-feature labels

Usability - "Tell me why I cannot open the image tool."

New

JL

Result explanation was useful

Review - Rated 5 out of 5

Reviewed

Selected feedback

Some evidence text is difficult to read

"The main result is clear, but the evidence summary and source name are smaller than I expected. Please make them easier to read."

Category

Accessibility

Submitted

24 Aug 2026, 9:12 AM

User tier

Free

Status

New

Mark reviewed

Add note

6.5 Data Engineer

Data Engineer Dashboard

uiabo.
Operational Portal

- Dashboard
- RPT Incorrect results
- PLN Pipeline monitoring
- My profile

Data Engineer Interface sample NO

DATA QUALITY OPERATIONS
Pipeline and report overview
Investigate reported results and monitor evidence-data updates.

Open incorrect-result reports
18
5 high priority

Reports reviewed today
7
+2 from yesterday

Healthy pipelines
5/6
1 delayed

Evidence freshness
96%
Within target

View pipeline health

Incorrect-result reports View report

REPORT	RESULT TYPE	CONCERN	REPORTED	STATUS
RPT-1048 Result may use outdated evidence	Text	High	24 Aug, 8:48 AM	New
RPT-1047 Source link unavailable	Link	Medium	23 Aug, 4:20 PM	Reviewing
RPT-1046 Explanation unclear	Text	Low	23 Aug, 2:03 PM	Resolved

Pipeline health 1 delayed

- FC Fact-check ingestion
Last update: 9:05 AM Healthy
- TS Trusted-source updates
Expected 8:30 AM Delayed
- EV Evaluation records
Last update: Yesterday Healthy

Open monitoring

Nora Ong
Data Engineer

Incorrect Result Report

uiabo.
Operational Portal

- Dashboard
- RPT Incorrect results
- PLN Pipeline monitoring
- My profile

Incorrect-result report Demo data NO

RPT-1048
Review reported analysis result
Investigate the result, evidence and recorded pipeline information.

Assign to me **Update status**

User report New · High priority

"I believe the announcement used in this result may be outdated. A newer official page appears to say something different."

Reported: 24 Aug 2026, 8:48 AM
Reason: Evidence may be outdated
Result ID: res_demo_1048
Input type: Short-form text

Original result High Concern - 82/100

Identified claim
"Singapore is introducing a new \$500 tax next week."

Original explanation
The available official announcement directly contradicted the submitted claim.

Evidence and provenance 1 source

SOURCE	PUBLISHED	RETRIEVED	STANCE
Example official announcement example.gov/announcement	10 Aug 2026	20 Aug 2026	Contradicting

Investigation status
New

Investigation notes
Record findings and follow-up actions

Save notes

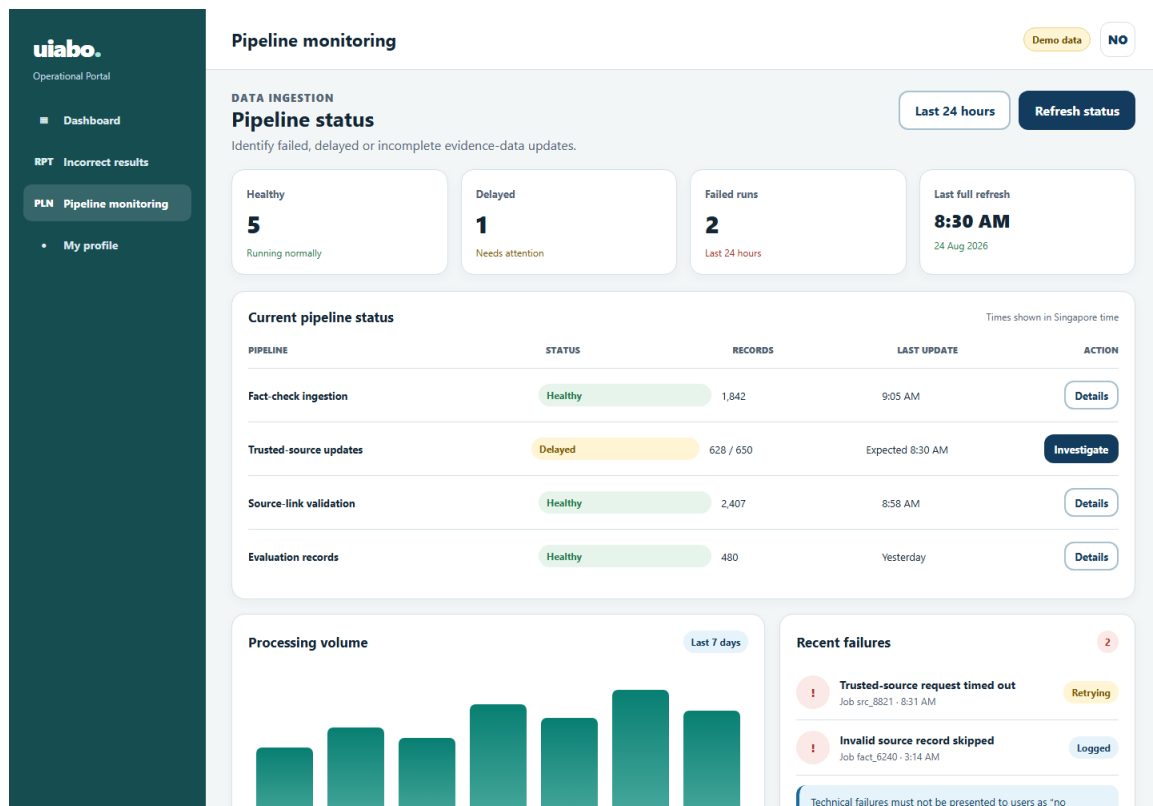
Pipeline record

- Version: sprint-1-v1
- Completed: 20 Aug, 10:00 AM
- Uncertainty: Medium
- Warnings: One source found

Privacy: view only the information required to investigate this report.

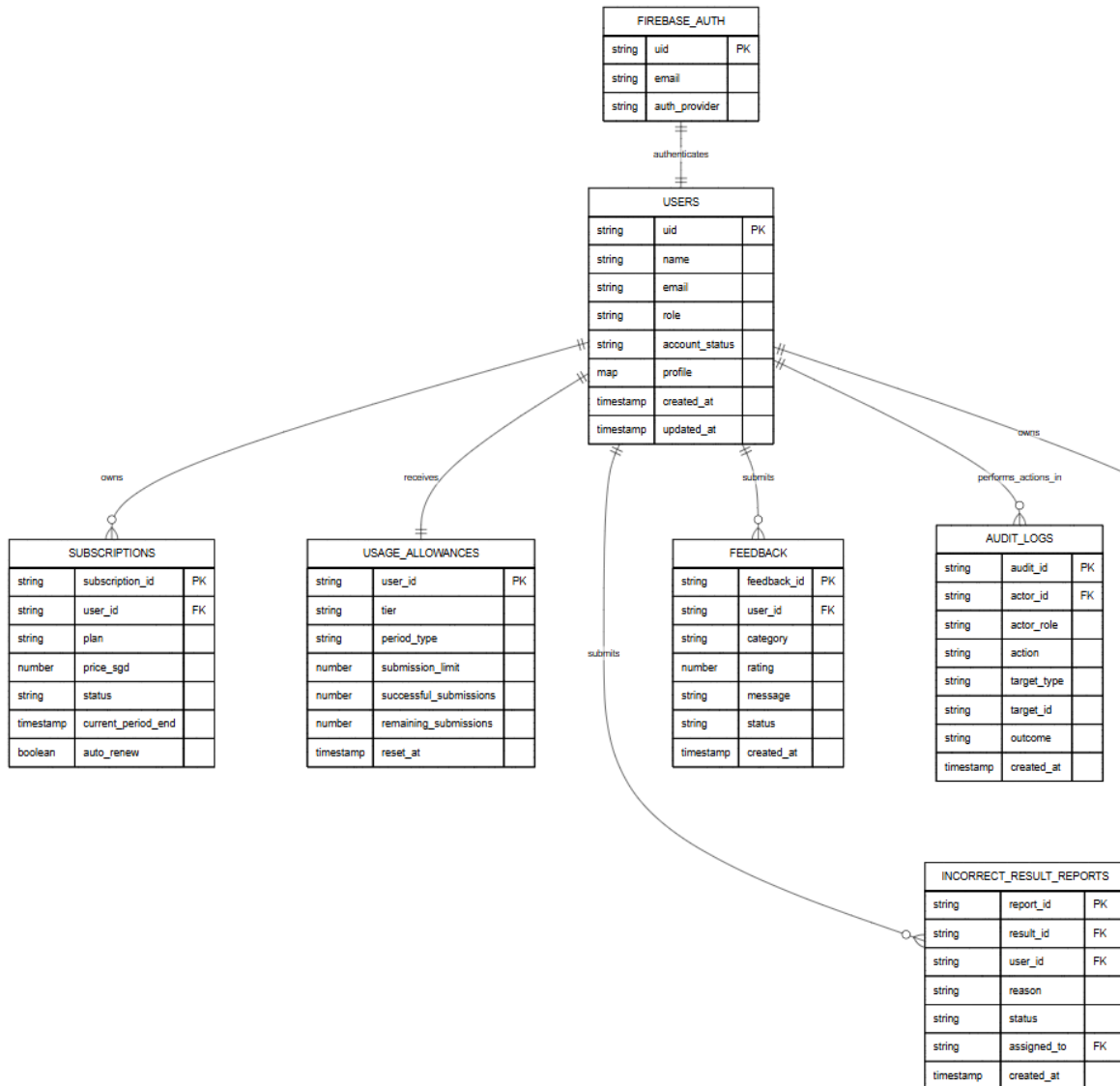
Nora Ong
Data Engineer

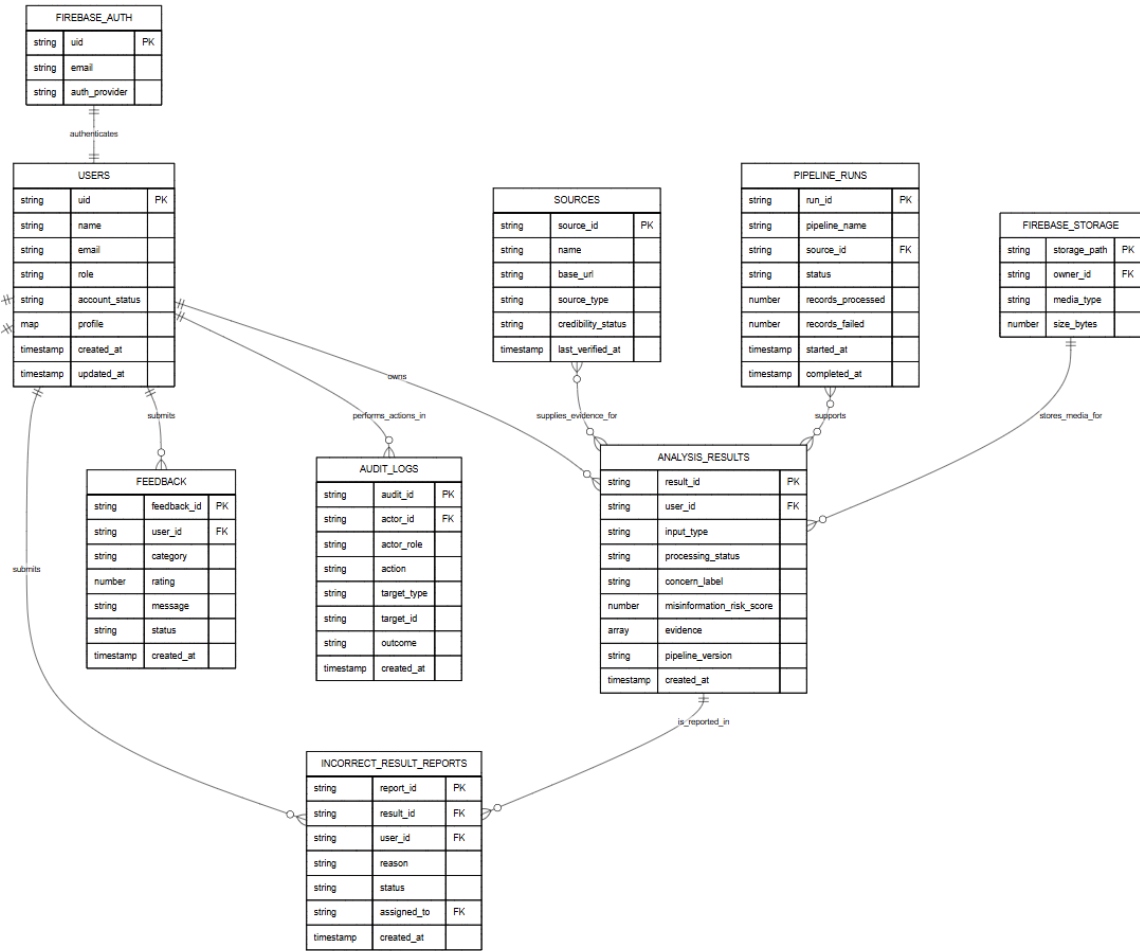
Pipeline Monitoring



7. Database Design

Mermaid Diagram





Cloud Firestore is a NoSQL document database. The lines in the diagram show document-ID references rather than SQL joins. Evidence snapshots are embedded inside analysis_results to match the current pipeline implementation and preserve the evidence used for each result.

7.1. Users Collection

Document path: users/{uid}

Stores the Firestore profile for every Free User, Premium User, System Administrator and Data Engineer. The document ID is the Firebase Authentication UID. Passwords are not stored here.

Field	Type	Required	Purpose
uid	String	Yes	Firebase Authentication UID and Firestore document ID.
name	String	Yes	User's display name.
email	String	Yes	Email copied from Firebase Authentication for profile display and authorised account queries.
role	String	Yes	Access role: free, premium, system_admin or data_engineer.
account_status	String	Yes	Account state: active, suspended or deactivated.
profile	Map	Yes	Editable profile information. Only data needed by the application should be stored.

Field	Type	Required	Purpose
storage_used_bytes	Number	Yes	Current personal storage usage used for the 80% warning.
storage_quota_bytes	Number	Yes	Maximum personal storage allocation.
last_login_at	Timestamp	No	Most recent successful login time.
created_at	Timestamp	Yes	When the profile record was created.
updated_at	Timestamp	Yes	When the profile record was last changed.

7.2. Subscriptions Collection

Document path: subscriptions/{subscription_id}

Stores premium subscription periods and renewal state. Payment-card details must remain with the payment provider and must not be stored in Firestore.

Field	Type	Required	Purpose
subscription_id	String	Yes	Unique subscription document ID.
user_id	String	Yes	UID of the user who owns the subscription.
plan	String	Yes	Plan identifier, such as premium_monthly.
price_sgd	Number	Yes	Recorded monthly price; SGD 20 for the prototype.
status	String	Yes	pending, active, cancelled, expired or payment_failed.
current_period_start	Timestamp	Yes	Start of the current paid period.
current_period_end	Timestamp	Yes	End of the current paid period.
auto_renew	Boolean	Yes	Whether automatic renewal remains enabled.
payment_provider	String	Yes	Provider or app-store mechanism used for payment.
provider_reference	String	No	Non-sensitive provider transaction or subscription reference.
created_at	Timestamp	Yes	When the subscription record was created.
updated_at	Timestamp	Yes	When the subscription record was last updated.

7.3. Usage Allowances Collection

Document path: usage_allowances/{user_id}

Stores the current free daily or premium monthly allowance. One successful Free User submission is allowed per day; Premium Users receive up to 60 successful submissions per month.

Field	Type	Required	Purpose
user_id	String	Yes	User UID and document ID.
tier	String	Yes	free or premium.
period_type	String	Yes	daily for free access or monthly for premium access.
period_start	Timestamp	Yes	Beginning of the current allowance period.
period_end	Timestamp	Yes	End of the current allowance period.
submission_limit	Number	Yes	Maximum successful submissions for this period: 1 or 60.
successful_submissions	Number	Yes	Number of completed analyses charged during the period.

Field	Type	Required	Purpose
remaining_submissions	Number	Yes	Remaining amount displayed in the application.
reset_at	Timestamp	Yes	When the allowance will reset.
updated_at	Timestamp	Yes	When the counter was last changed.

7.4. Analysis Results Collection

Document path: analysis_results/{result_id}

Stores completed and failed pipeline records. Result history is retrieved by querying user_id and created_at. Evidence is embedded as an array so the explanation preserves the evidence that was used when the result was generated.

Field	Type	Required	Purpose
result_id	String	Yes	Unique analysis result and document ID.
user_id	String	Yes	UID of the user who owns the result.
input_type	String	Yes	text, link, image_caption, ocr, context or deepfake.
original_text	String	No	Submitted text, caption or corrected OCR text.
submitted_url	String	No	Public webpage URL for a link analysis.
media_storage_path	String	No	Firebase Storage path for an authorised image submission.
processing_status	String	Yes	completed or failed.
extracted_claim	String	No	Factual claim identified by the pipeline.
claim_category	String	Yes	factual, opinion, satire, prediction, personal experience or unverifiable.
checkable	Boolean	Yes	Whether the identified claim can be fact-checked.
concern_label	String	No	Low Concern, Needs Caution, High Concern or Not Enough Information.
misinformation_risk_score	Number	No	Risk indicator from 0 to 100; null when information is insufficient.
ai_manipulation_likelihood	Number	No	Premium image AI-generation or manipulation likelihood from 0 to 100.
uncertainty	String	No	Low, Medium or High uncertainty.
uncertainty_reasons	Array	No	Plain-language reasons for uncertainty.
explanation	String	No	Plain-language result explanation.
recommended_action	String	No	Suggested next action for the user.
evidence	Array<Map>	No	Evidence snapshots containing source, URL, passage, stance and quality score.
warnings	Array	No	Non-fatal preparation, retrieval or assessment warnings.
pipeline_version	String	Yes	Version of the pipeline that generated the record.
failure	Map	No	Failure stage, error code, safe message and retryable flag for failed runs.
share_token	String	No	Random token used when result sharing is enabled.
share_enabled	Boolean	Yes	Whether a shareable result link is active.
created_at	Timestamp	Yes	When analysis processing finished or failed.
saved_at	Timestamp	Yes	Server timestamp written by Firestore.

7.5. Incorrect Result Reports Collection

Document path: incorrect_result_reports/{report_id}

Stores reports submitted from an analysis result and the Data Engineer's investigation state.

Field	Type	Required	Purpose
report_id	String	Yes	Unique report document ID.
result_id	String	Yes	Analysis result being reported.
user_id	String	Yes	UID of the reporting user.
reason	String	Yes	Selected reason or report category.
user_comment	String	No	Optional explanation supplied by the user.
status	String	Yes	new, investigating, resolved or rejected.
priority	String	Yes	low, medium or high investigation priority.
assigned_to	String	No	UID of the assigned Data Engineer.
investigation_notes	Array<Map>	No	Timestamped investigation notes and permitted status updates.
resolution_summary	String	No	Final investigation outcome without unnecessary personal information.
created_at	Timestamp	Yes	When the user submitted the report.
updated_at	Timestamp	Yes	When the report was last changed.
resolved_at	Timestamp	No	When the investigation was closed.

7.6. Feedback Collection

Document path: feedback/{feedback_id}

Stores application reviews, accessibility feedback, bug reports and general comments for System Administrator review.

Field	Type	Required	Purpose
feedback_id	String	Yes	Unique feedback document ID.
user_id	String	Yes	UID of the user who submitted the feedback.
category	String	Yes	review, accessibility, bug or general.
rating	Number	No	Optional service rating, normally from 1 to 5.
message	String	Yes	Feedback text entered by the user.
status	String	Yes	new, reviewed or resolved.
reviewed_by	String	No	UID of the reviewing System Administrator.
admin_note	String	No	Operational follow-up note.
created_at	Timestamp	Yes	When the feedback was submitted.
updated_at	Timestamp	Yes	When the feedback record was last changed.

7.7. Sources Collection

Document path: sources/{source_id}

Stores trusted-source and provenance metadata used by evidence retrieval and maintained by authorised Data Engineers.

Field	Type	Required	Purpose
source_id	String	Yes	Unique source document ID.
name	String	Yes	Publisher or organisation name.
base_url	String	Yes	Canonical source website URL.
source_type	String	Yes	fact_check, government, academic, news or other.
credibility_status	String	Yes	trusted, review_required or blocked.
country	String	No	Country or region associated with the source.
provenance_notes	String	No	Reason for its status and relevant validation notes.
active	Boolean	Yes	Whether the retrieval pipeline may currently use the source.
last_verified_at	Timestamp	Yes	Most recent source validation time.
created_at	Timestamp	Yes	When the source record was created.
updated_at	Timestamp	Yes	When source metadata was last changed.

7.8. Pipeline Runs Collection

Document path: pipeline_runs/{run_id}

Stores the monitoring data needed to identify running, delayed, incomplete and failed ingestion updates.

Field	Type	Required	Purpose
run_id	String	Yes	Unique pipeline run document ID.
pipeline_name	String	Yes	Name of the ingestion or update pipeline.
source_id	String	No	Related source record when the run imports source data.
status	String	Yes	running, completed, delayed, incomplete or failed.
records_expected	Number	No	Expected number of records, when known.
records_processed	Number	Yes	Number of successfully handled records.
records_failed	Number	Yes	Number of failed records.
data_range	Map	No	Affected dates, record range or dataset version.
error_summary	String	No	Safe summary of the run failure or delay.
triggered_by	String	Yes	system or authorised operational-user UID.
started_at	Timestamp	Yes	When the run started.
completed_at	Timestamp	No	When the run ended.
updated_at	Timestamp	Yes	Most recent monitoring update.

7.9. Audit Logs Collection

Document path: audit_logs/{audit_id}

Stores immutable records of sensitive operational actions such as account queries, account views, account creation, suspension, unsuspension and investigation updates.

Field	Type	Required	Purpose
audit_id	String	Yes	Unique audit record ID.
actor_id	String	Yes	UID of the operational user or system process.
actor_role	String	Yes	system_admin, data_engineer or system.
action	String	Yes	Action that was attempted.
target_type	String	Yes	Collection or resource type affected.

Field	Type	Required	Purpose
target_id	String	No	Identifier of the affected record.
reason	String	No	Required reason for sensitive changes where applicable.
outcome	String	Yes	success, denied or failed.
metadata	Map	No	Non-sensitive context needed for security or investigation.
created_at	Timestamp	Yes	Server timestamp. Audit records are not edited after creation.

7.10. Supporting Firebase Services

These services support the database but are not Firestore collections.

Service	Responsibility
Firebase Authentication	Stores email-and-password credentials, creates the UID and manages authenticated sessions. Password hashes must not be copied into Firestore.
Firebase Storage	Stores uploaded images and screenshots. Firestore keeps only authorised storage paths and media metadata.
Cloud Firestore	Stores the document collections described above. Security rules and backend role checks restrict every read and write.